

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #10

March 23 - 27, 2026

Instructor: J. Michael Ruohoniemi

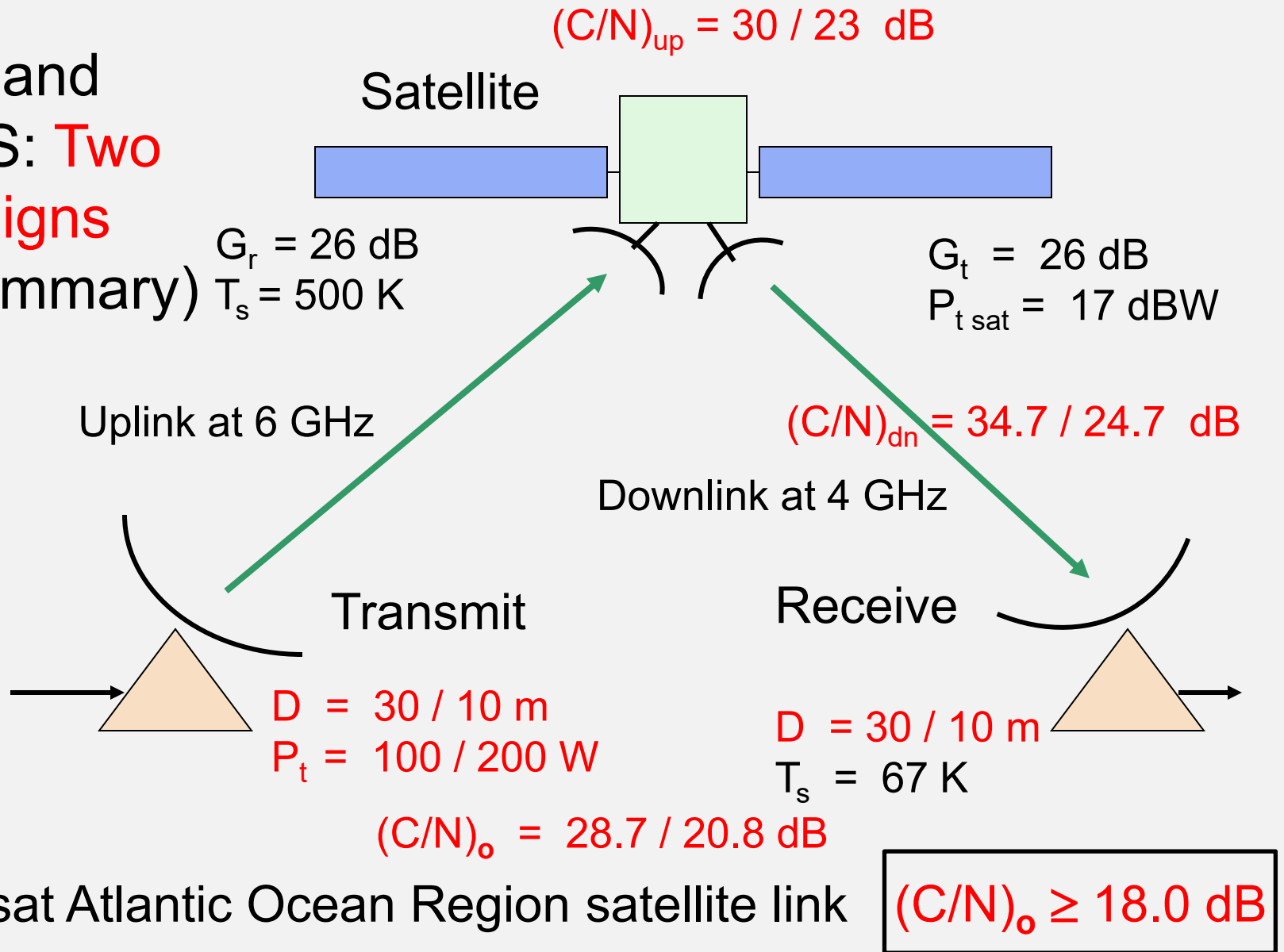
Topics (1/2)

- Mid-term de-brief
- Intelsat C-band system: One-slide review
- Link margins for rain
- Direct Broadcast Satellite Television (DBS-TV)
- DBS-TV: Analysis of link margin

Topics (2/2)

- Course Project Preliminaries
- Modulation / Analog signal / FM modulation
- Digital Tx and Rx of Analog signals
- Sampling Theorem / Sampling
- Analog to Digital Conversion / Pulse Code Modulation / Quantization

C-band
FSS: **Two**
designs
(Summary)



Intelsat Atlantic Ocean Region satellite link

Rain Attenuation Margins

- The values of overall C/N for the two designs were obtained assuming clear air
- When rain affects the links, C/N values will be reduced
- Need to re-calculate uplink and downlink margins
- A *margin* is the amount by which a value can be reduced before hitting a limiting condition
- Limiting condition is still $(C/N)_o = 18.0 \text{ dB}$

Rain Attenuation Margins

- Question:

How much can $(C/N)_{up}$ or $(C/N)_{dn}$ be reduced before $(C/N)_o = 18.0$ dB ?

- That is, how much rain can we tolerate on either link?
- We will discuss the amount of rain in terms of A dB, the total attenuation on the signal due to rain on a path
- With a rain model we can relate A dB to rain type, rate, and path length

Rain Attenuation Margins

- We can assume that the occurrence of heavy rain on the uplink and downlink are independent events
- If the uplink is affected by heavy rain, assume downlink is in clear air and vice-versa
- (Not true for a *loop-back test*)

Uplink Rain Attenuation Margin

- Uplink rain attenuation A_{up} dB:

- Power received at transponder input is

$$P_r \text{ rain} = P_r \text{ clear air} - A_{up} \text{ dBW}$$

$$(C/N)_{up} \text{ rain} = (C/N)_{up} - A_{up} \text{ dB}$$

- Assuming linear transponder operation

$$P_t \text{ rain} = P_t \text{ clear air} - A_{up} \text{ dBW}$$

$$(C/N)_{dn} \text{ rain} = (C/N)_{dn} - A_{up} \text{ dB}$$

Uplink Rain Attenuation Margin

- Note that this result is true of a *linear* transponder for which P_t decreases by the same amount as P_r
- For a *non-linear* transponder the decrease in P_t will generally be less
- For a *regenerative* transponder, or one that utilizes automatic gain control (AGC), P_t might not drop at all

Uplink Rain Attenuation Margin

- For a linear transponder the C/N on both the uplink and downlink is reduced by A_{up}
- This can be worked through the reciprocal formula to show that $(C/N)_o$ is reduced by A_{up}
- Conclusion:

With rain on the uplink and a linear transponder $(C/N)_o$ is reduced by exactly A_{up} dB (the uplink rain attenuation)

Uplink Rain Attenuation Margin

- The uplink rain attenuation margin is

$$\text{Uplink margin} = (C/N)_o \text{ clear air} - (C/N)_{o \text{ min}}$$

- In the C-band link example with the smaller 10 m antennas

$$(C/N)_o \text{ clear air} = 20.8 \text{ dB}, (C/N)_{o \text{ min}} = 18.0 \text{ dB}$$

- Uplink rain attenuation margin is 2.8 dB (~ 3 dB)
- Whether this is enough will depend on rainfall conditions

Uplink Rain Attenuation Margin

- For the uplink, the margin on rain attenuation is the same as the margin on $(C/N)_o$ – for a linear transponder
- If we can tolerate a drop of 3 dB in $(C/N)_o$ then we can tolerate rain attenuation of 3 dB on the uplink
- This is not the case for the downlink – the margin on rain attenuation will be less than the margin on $(C/N)_o$

Downlink Rain Attenuation Margin

- For the downlink, there is (of course) direct attenuation of the power in the carrier wave
- But there is also a rise in noise power in the receiver because rain on the downlink increases the sky noise temperature
- These two factors generally lower the C/N on the downlink more than would be the case for the same rain attenuation on the uplink

Downlink Rain Attenuation Margin

- The overall link is typically more vulnerable to rain on the downlink
- Next, we will apply our noise model for a lossy medium to determine the rise in noise power, the resulting value of $(C/N)_{dn}$, the impact on $(C/N)_o$, and the margin for rain attenuation on the downlink

Downlink C/N Margin

- Downlink rain attenuation A_{dn} dB:

$$(C/N)_{dn \text{ rain}} = (C/N)_{dn \text{ clear air}} - \Delta(C/N)_{dn}$$

- Power received at transponder input is same as in clear air

$$(C/N)_{up} = (C/N)_{up \text{ clear air}}$$

- To find the downlink rain attenuation margin we must rearrange the reciprocal formula

$$1/(C/N)_{dn \text{ min}} = 1/(C/N)_{o \text{ min}} - 1/(C/N)_{up}$$

Downlink C/N Margin

- In the C-band link example with 10 m antennas:

$$(C/N)_{o \min} = 18.0 \text{ dB}$$

$$(C/N)_{up \text{ clear air}} = 23.0 \text{ dB}$$

$$(C/N)_{dn \text{ clear air}} = 24.7 \text{ dB}$$

$$(C/N)_{dn \min} = ?$$

- $$\begin{aligned} 1/(C/N)_{dn \min} &= 1/(C/N)_{o \min} - 1/(C/N)_{up} \\ &= 1 / 63.1 - 1 / 200 = 92.2 \end{aligned}$$

$$(C/N)_{dn \min} = 19.6 \text{ dB}$$

- Downlink clear air C/N margin is $24.7 - 19.6 =$
5.1 dB

C-band Link Margins

- At C-Band rain attenuation is small
- Uplink margin of ~ 3 dB at 6 GHz will not be exceeded even in heavy rain
- Downlink rain attenuation will be smaller for the same rain on account of lower frequency, < 2 dB
- Seems like we are better off on the downlink?
- Reduction in carrier power, C , in $(C/N)_{dn}$ is not the entire story!

C-band Link Margins

- Sky noise temperature *increases* because rain absorbs energy at 4 GHz
- An absorber is also a radiator – so sky noise increases as rain attenuation increases
- The noise power, N , in the receiver increases
- Leading to an additional reduction in $(C/N)_{dn}$

C-band Downlink Margins

- Example for 4 GHz downlink:
- Assume clear air attenuation is 0.1 dB with *output noise model* expression (and $T_p = 290^\circ \text{ K}$)

$$T_n = 290 (1 - G_l) = 290 (1 - 10^{-0.1/10}) = 7 \text{ K}$$

- $T_{\text{ant}} \approx T_{\text{sky}} = 7 \text{ K}$
- Assume $T_{\text{LNA}} = 60 \text{ K}$ with a high gain LNA
- Then $T_s = (\text{clear air}) \approx T_{\text{sky}} + T_{\text{LNA}} = 67 \text{ K}$

C-band Downlink Margins

- Rain attenuation of 1 dB causes path attenuation of 1.1 dB (rain + clear air)

- Gives sky noise temperature in rain

$$T_{\text{sky rain}} = 290 (1 - 0.776) = 64.9 \text{ K}$$

- New system noise temperature in rain is

$$\begin{aligned} T_{\text{s rain}} &\approx T_{\text{sky rain}} + T_{\text{LNA}} \\ &= 64.9 + 60 = 124.9 \text{ K} \end{aligned}$$

- Increase in noise power in receiver is

$$\Delta N = 10 \log (124.9 / 67) = 2.7 \text{ dB}$$

C-band Downlink Margins

- In the rain we have an additional 1 dB of attenuation along the path and an increase in noise power of 2.7 dB
- Then $(C/N)_{dn}$ for 1 dB of rain attenuation along the path is
$$\begin{aligned} &= (C \text{ clear air} - A_{dn}) / (N \text{ clear air} + \Delta N) \\ &= C/N \text{ clear air} - A_{dn} - \Delta N \\ &= 24.7 - 1.0 - 2.7 = 21.0 \text{ dB} \quad (\text{ref: } 24.7 \text{ dB}) \end{aligned}$$

C-band Downlink Margins

- 1 dB of rain attenuation on the downlink has caused a 3.7 dB drop in $(C/N)_{dn}$

$$(C/N)_{dn \text{ rain}} = 21.0 \text{ dB}$$

- Recall $(C/N)_{dn \text{ min}} = 19.6 \text{ dB}$
- We still have margin to work with, 1.4 dB on downlink (C/N)

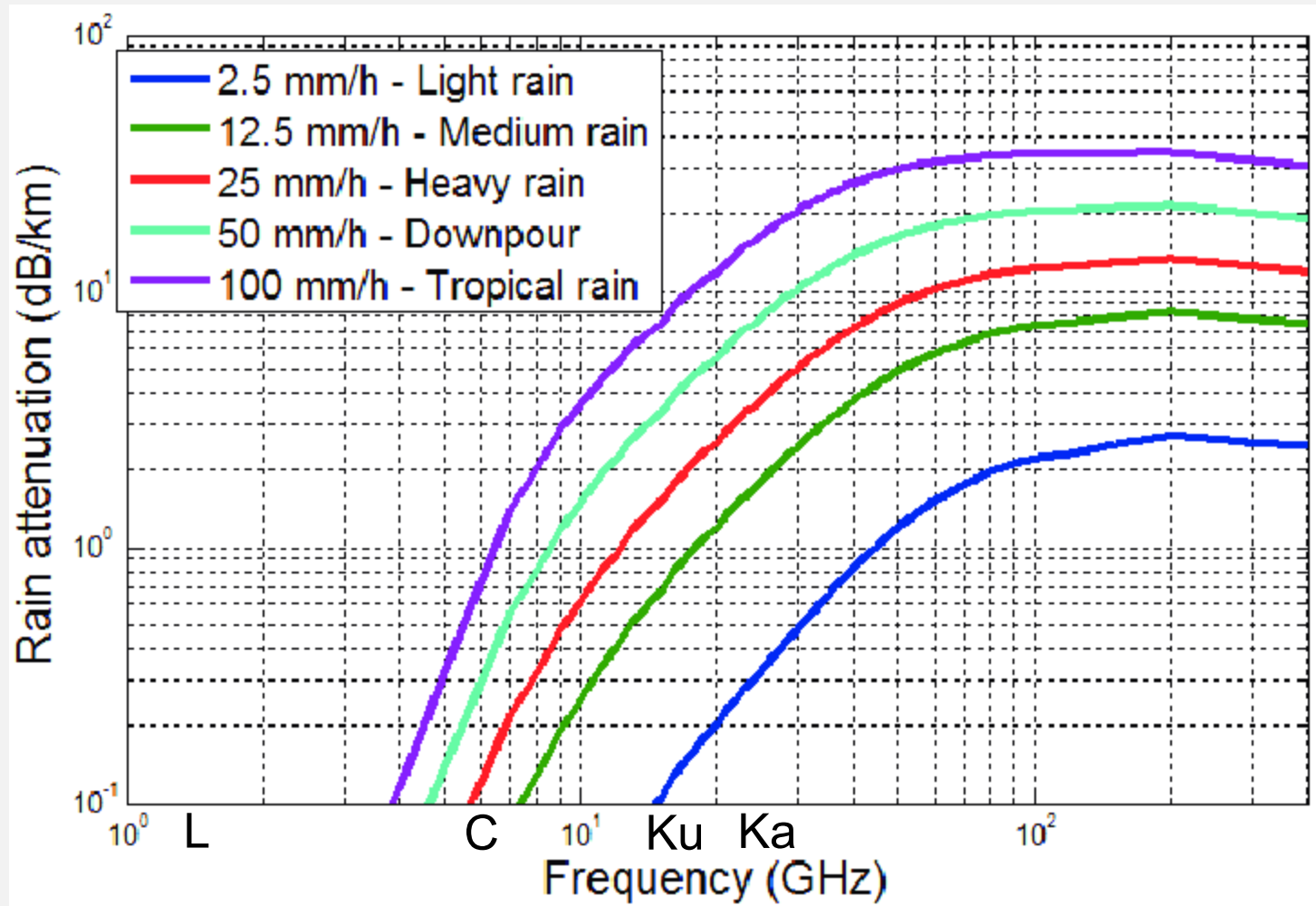
C-band Downlink Margins

- Observe that 1 dB of direct rain attenuation used up 3.7 dB of margin on the downlink
- We can iterate on the rain attenuation, increasing it until all the margin is gone
- This will establish the margin for rain attenuation on the downlink
- In this example, the tolerance for rain on the downlink (< 2 dB) will be less than that on the uplink (~ 3 dB)

Rain Attenuation Summary

- Downlinks are generally more susceptible to outages from rain because of (i) lower transmit power at the satellite and (ii) the sky noise factor
- Satcom design requires consideration of weather factors, especially at the higher frequencies

Rain Attenuation as a Function of Frequency



Source: ResearchGate

Direct Broadcast Satellite Television

- Direct Broadcast Satellite (DBS) has been a big success story of satellite communications
- Video distribution at one time earned more than half of all satellite communications revenues
- 33 M customers in US in 2017, more in Europe
- What satellites do best:
point to multipoint = broadcasting

Direct Broadcast Satellite Television

- Satellite internet is often coupled with DBS
- Main vendors (US):
 - Directv (spun off by AT&T in 2021)
 - EchoStar (merged with DISH Network in 2024)
- Often the only choice for rural users
- Growth in subscribers flattened out by 2015 as internet TV service became available
- DBS television is now threatened by alternatives

DBS TV – A Historical Note on C-band

- In the 1980s C-band satellite systems for video distribution began to be tapped by home users – the signal was unencrypted
- Required ~ 3 m antennas (rather large)
- Rather than prosecute users, Congress required that companies encrypt their signal
- Some free signal is still available on C-band, and there are enthusiasts

C-band Satellite Dish, aka “Big Ugly Dish”



Direct Broadcast Satellite Television

- Types of satellites:

- Directv

Boeing 702

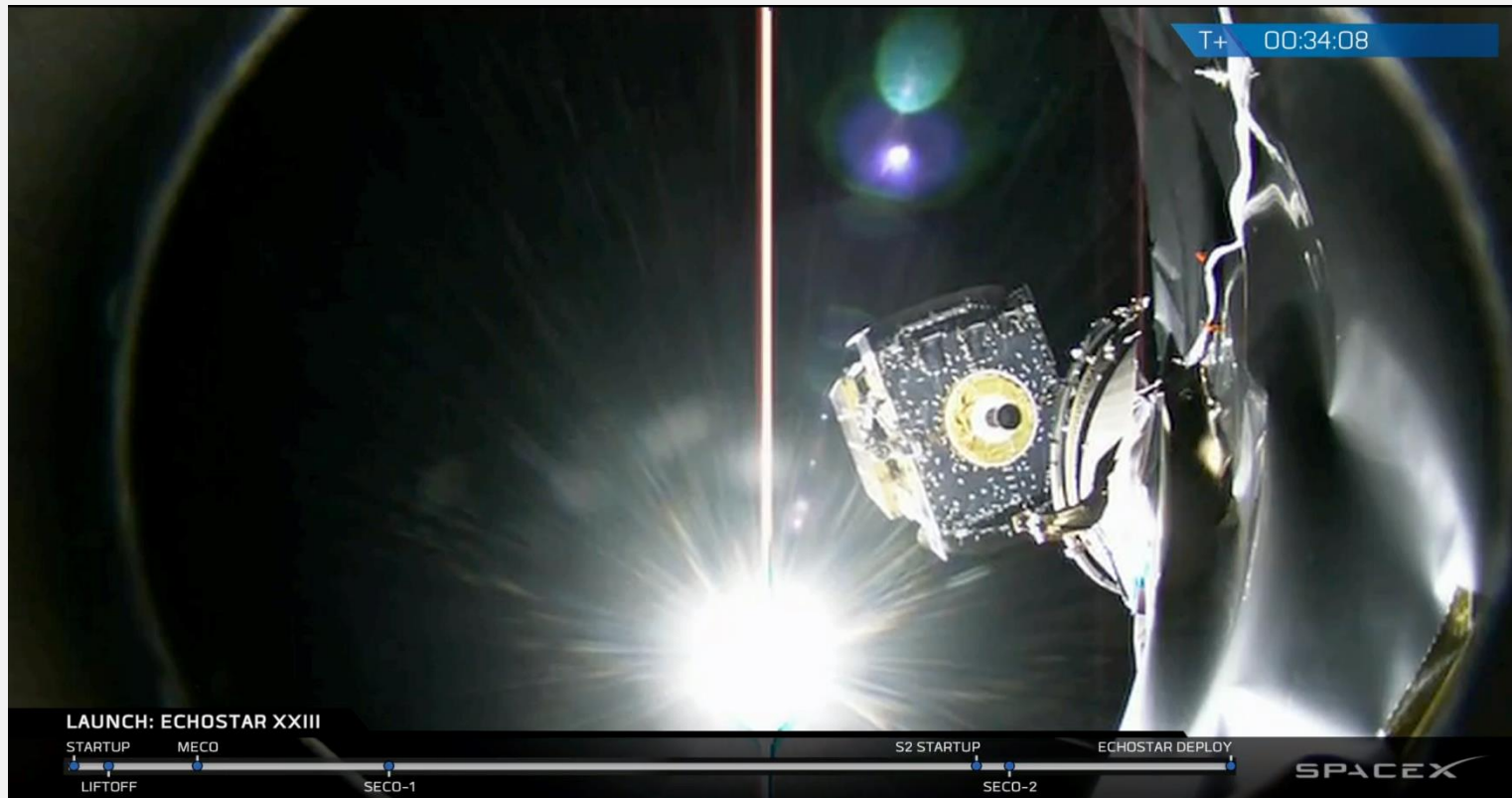
Three axis stabilized satellite

- Echostar

Space Systems/Loral 1300 spacecraft bus

- SpaceX delivered Echostar XXIII into orbit in 2017 using the Falcon 9 booster

View of the EchoStar XXIII satellite as it deployed from the Falcon 9 second stage



Artist's depiction
of EchoStar XXIII
deployed in orbit



DBS-TV Economics

System Design Philosophy:

High power uplink

High power satellite

Small receiving terminal

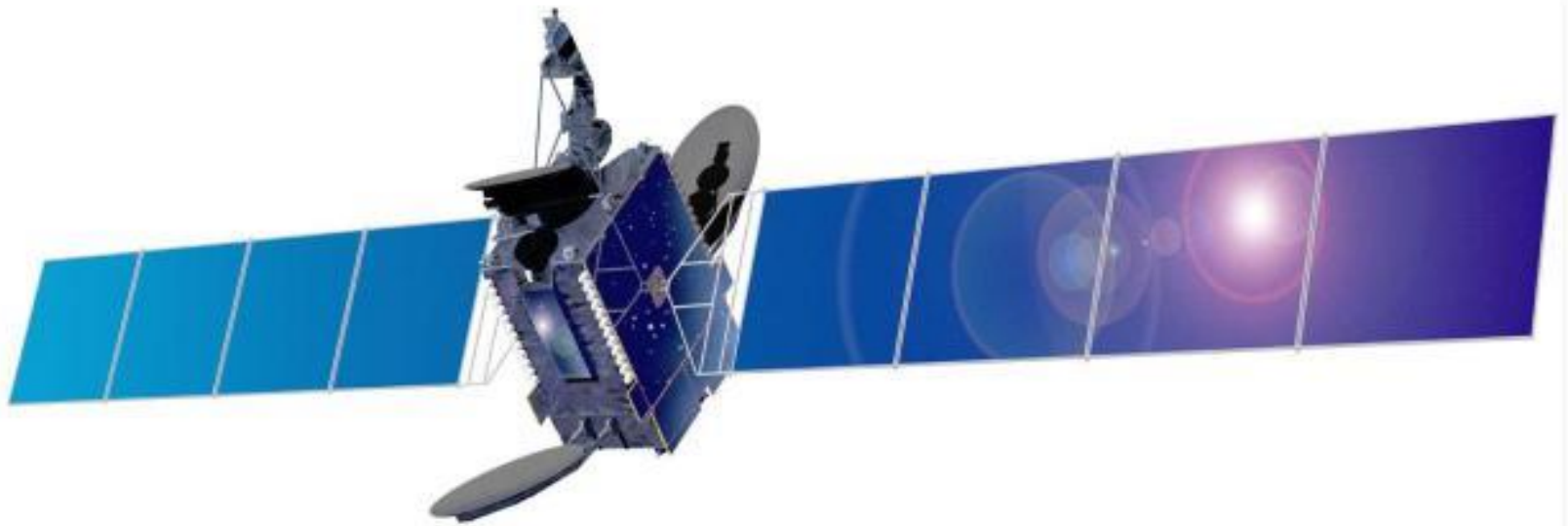
- Key improvement over earlier C-band systems was the development of low-cost Ku-band (small) antennas and receivers

Echostar DBS TV Satellite

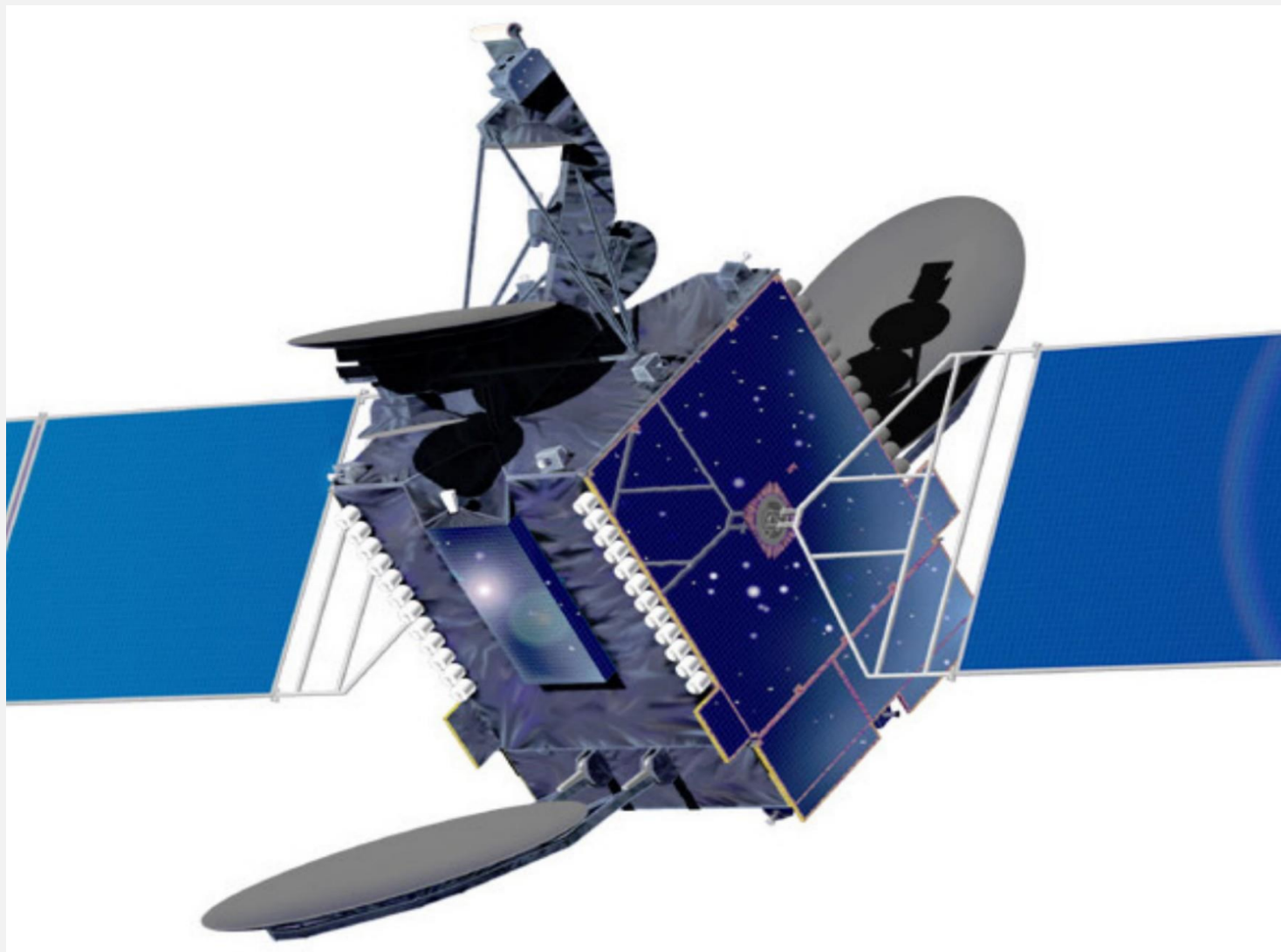
- First launch of an EchoStar satellite was in 1995
- Currently there are 10 EchoStar satellites operating between 61.5° W and 119° W
- Coverage:
 - Continental U.S., Hawaii, Alaska and P-R
- Due to declining subscriptions the company is saddled with billions in debt
- An attempt to merge with DirecTV was called off
- Sales of spectrum and spectrum wars are underway

Early versions – Echostar 6 DBS TV Satellite

- Satellite Manufacturer: Space Systems / Loral
- Satellite type: SSL 1300
- Overall length: 102.1 ft. (31.1 m)
- Launch weight: 8157 lbs. (3,700 kgs)
- Beginning of life power: 11.27 kW
- Batteries: Nickel Hydrogen
- Eclipse capability: 100%



Echostar 5



Echostar 6 DBS TV Satellite

- Reflector Antennas:
1 Receive (47") 2 Transmit (94")
- Transponders: 32
- Number of digital TV channels available:
Approx. 150
- DBS TV band is 12.2 to 12.7 GHz (Ku band)
(Exclusive allocation because of high power)

DBS TV Satellite

- Transponder saturated output power = 160 W = 22.0 dBW, apply 1.5 dB backoff, $P_t = 20.5$ dBW
- Maximum transmit antenna gain = 34.3 dB
- EIRP = $20.5 + 34.3 = 54.8$ dBW
- Path loss = 206.0 dB (typical)
- Clear air atmospheric loss is 0.4 dB
- Miscellaneous loss is 0.4 dB
- System is designed for > 99.7% availability
- 0.3% outage time per year ~ 24 hours / year



Fig 10.5 - text DBS-TV receive antennas

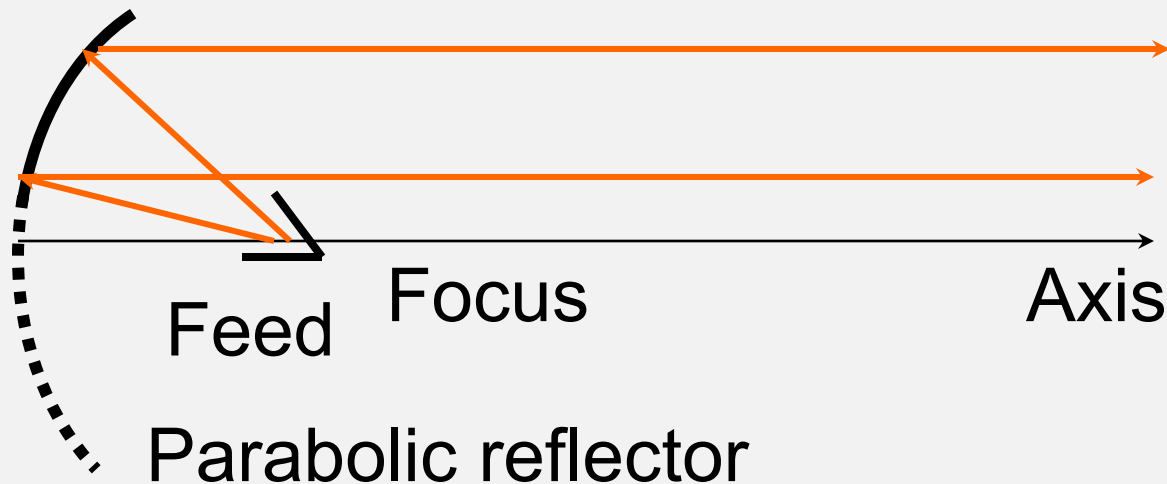
DBS-TV Receiving Stations

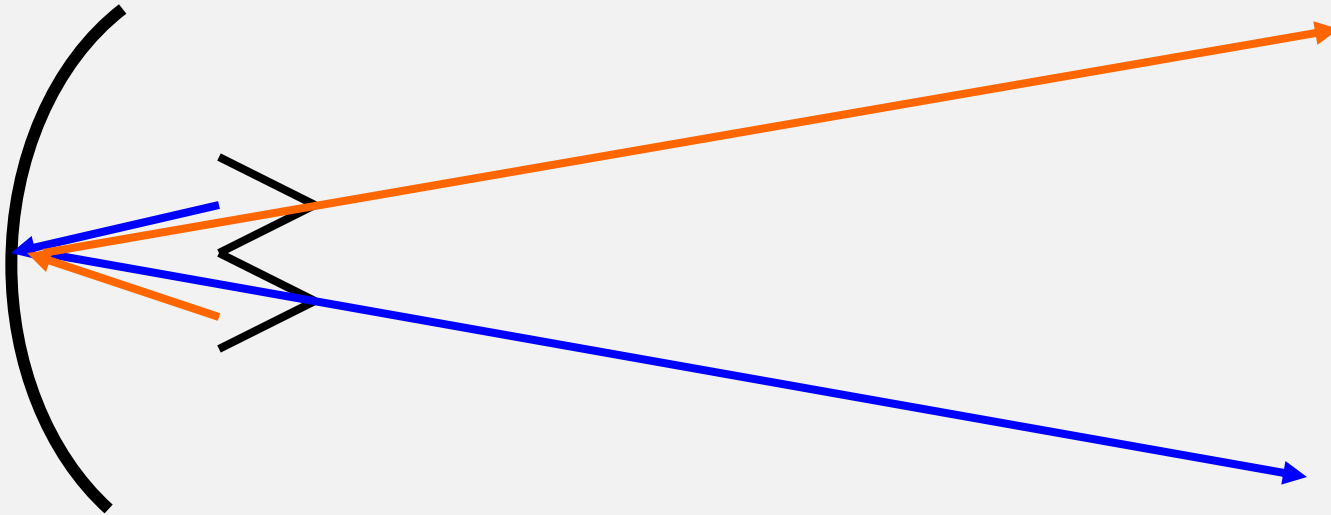
- Utilizes small, 18 inch to 24 inch, Ku-band dish antennas
- High efficiency offset-fed design, $\eta = 70\%$

$$G = \eta (\pi D / \lambda)^2 = 0.7 (\pi 0.457 / 0.024)^2 \\ = 2505 \text{ or } 34.0 \text{ dB}$$

Offset Dish Antennas

- Offset reflector has less blockage of the antenna by the feed antenna and its supports
- Leads to higher aperture efficiency





Dual offset feeds with a flat dish give two beams in azimuth; beamwidth sets the angle of minimum separation of satellites

DBS-TV Downlink Budget

- Downlink power budget for receiver on
 - 3 dB contour of satellite beam follows on next slide

(Ref: Table 4.6a in text)

Parameters are typical for DBS-TV systems
– not specific to Directv or Echostar systems

DBS-TV Downlink Budget

- | | |
|------------------------------|-----------------|
| EIRP | 54.8 dBW |
| G_r receiving terminal | 33.8 dB |
| Downlink path loss (typical) | - 206.0 dB |
| Edge of beam loss | - 3.0 dB |
| Miscellaneous losses | -0.4 dB |
| Clear air atmospheric loss | <u>- 0.4 dB</u> |
| $P_r =$ | - 121.2 dBW |

DBS -TV Downlink C/N

- Noise power in earth station receiver

$$k \quad -228.6 \text{ dBW}$$

$$T_s = 110 \text{ K} \quad 20.4 \text{ dBK}$$

$$B_N = 20 \text{ MHz} \quad \underline{73.0 \text{ dBHz}}$$

$$N \quad -135.2 \text{ dBW}$$

- Hence

$$\begin{aligned} C/N &= -121.2 \text{ dBW} - 135.2 \text{ dBW} \\ &= 14.0 \text{ dB} \end{aligned}$$

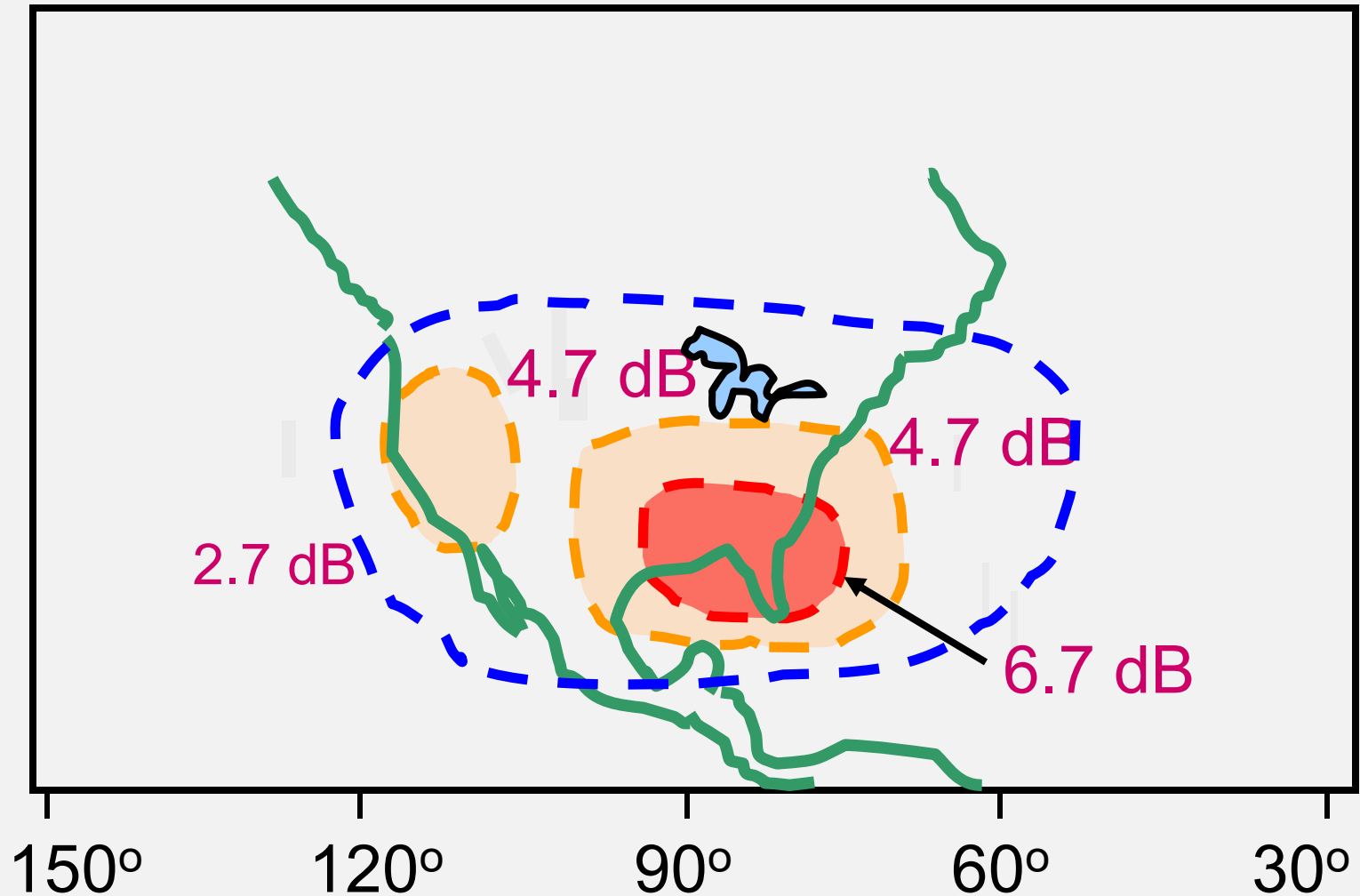
DBS TV Satellite

- C/N down = 14.0 dB
- Determine C/N overall
- C/N up ?
- Set C/N up so high that it does not limit the C/N overall
- With C/N up > 20 dB, C/N overall is practically the same as C/N down, ~14.0 dB
- In fact, C/N up $\geq$ 34 dB

DBS TV Satellite

- The threshold for operation is set at C/N overall = 8.3 dB
- The link margin overall is set by the link margin on the downlink
- Link margin (downlink) is $14.0 - 8.3 = 5.7$ dB for stations within the -3 dB contour of the satellite footprint (allowed in link budget)

DBS-TV Link Margins (after Fig. 10.14)



DBS TV Satellite

- The satellite antenna footprint covers the continental United States (Conus)
- Shaped to provide stronger signal where rain attenuation is highest – SE of US
- Inside the 2 dB contour the link margin (downlink) rises to 6.7 dB - allows at least 4 dB of rain attenuation and 99.7% availability
- Inside the 4 dB contour the link margin (downlink) is 4.7 dB – most of heavily populated US

Rain Attenuation

- The link margin for a typical DBS-TV downlink is 5.7 dB on the -3 dB contour of footprint
- What rain attenuation can we sustain on the downlink?
- Some of the 5.7 dB margin is allocated to attenuation, some to noise power increase
- Allocation process requires iteration (trial and error)

Rain Attenuation

- Let's start by calculating the increase in system noise temperature for 3 dB rain attenuation on the downlink
- Clear air attenuation = 0.4 dB
- Total path attenuation = 3.4 dB
- Sky noise temperature in rain is

$$T_{\text{sky rain}} = 270 (1 - 10^{-0.34}) = 147 \text{ K}$$

- Medium temperature 270 K is a best-fit case for Ku-band (versus 290 K for C band)

Rain Attenuation

- Check where we are in clear air conditions, atmospheric attenuation is 0.4 dB

$$T_{\text{sky}} = 270 (1 - 10^{-0.04}) = 24 \text{ K}$$

- T_{sky} has increased from 24 K to 147 K because of the rain
- Recall that T_s clear = 110 K, so we can deduce that $T_{\text{LNA}} \approx 86 \text{ K}$

Rain Attenuation

- Find the system noise temperature for 3 dB rainfall conditions
- With 100% coupling of sky noise to antenna output we have

$$T_{s \text{ rain}} \approx T_{\text{LNA}} + T_s = 86 + 147 \text{ K} = 233 \text{ K}$$

Rain Attenuation

- The increase in receiver noise power can be found as

$$\begin{aligned}\Delta N_{\text{rain}} &= 10 \log_{10} (233/110) \\ &= 3.3 \text{ dB}\end{aligned}$$

- Change in C/N is due to both 3 dB attenuation and a 3.3 dB increase in noise power:

$$(C/N)_{\text{dn rain}} = 14.0 - 3.0 - 3.3 = 7.7 \text{ dB}$$

- This is below the threshold value of 8.3 dB by 0.6 dB

Rain Attenuation

- Conclusion: we cannot tolerate 3 dB of rain attenuation on the downlink
- How much can we tolerate?
- Taking the iterative approach, we guess 2.5 dB and recalculate:

$$T_{\text{sky rain}} = 270 (1 - 10^{-0.29}) = 131.5 \text{ K}$$

$$T_{\text{s rain}} = T_{\text{LNA}} + T_{\text{s}} = 86 + 131.5 \text{ K} = 217.5 \text{ K}$$

$$\Delta N_{\text{rain}} = 10 \log_{10} (217.5/110) = 2.96 \text{ dB}$$

$$(C/N)_{\text{dn rain}} = 14.0 - 2.5 - 2.96 = 8.54 \text{ dB}$$

Rain Attenuation

- We are slightly (.24 dB) above the 8.3 dB threshold
- One more tweak, try 2.65 dB of attenuation:

$$T_{\text{sky rain}} = 270 (1 - 10^{-.305}) = 136.2 \text{ K}$$

$$T_{\text{s rain}} = T_{\text{LNA}} + T_{\text{s}} = 86 + 136.2 \text{ K} = 222.2 \text{ K}$$

$$\Delta N_{\text{rain}} = 10 \log_{10} (222.2/110) = 3.05 \text{ dB}$$

$$(C/N)_{\text{dn rain}} = 14.0 - 2.65 - 3.05 = 8.3 \text{ dB}$$

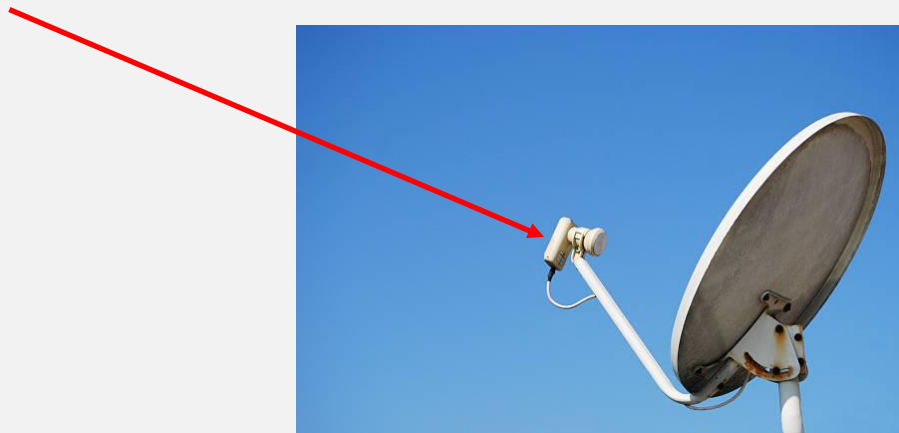
- Conclusion: we can tolerate 2.65 dB of rain attenuation on the -3 dB antenna contour

Rain Attenuation

- Noise power increases less rapidly as attenuation increases above 3.0 dB
- Stations on the -2 dB contour of the satellite footprint can sustain close to 4 dB rain attenuation
- How often does 3 dB rain attenuation occur in the US?
- Depends on location ... see Chapter 10 of text!

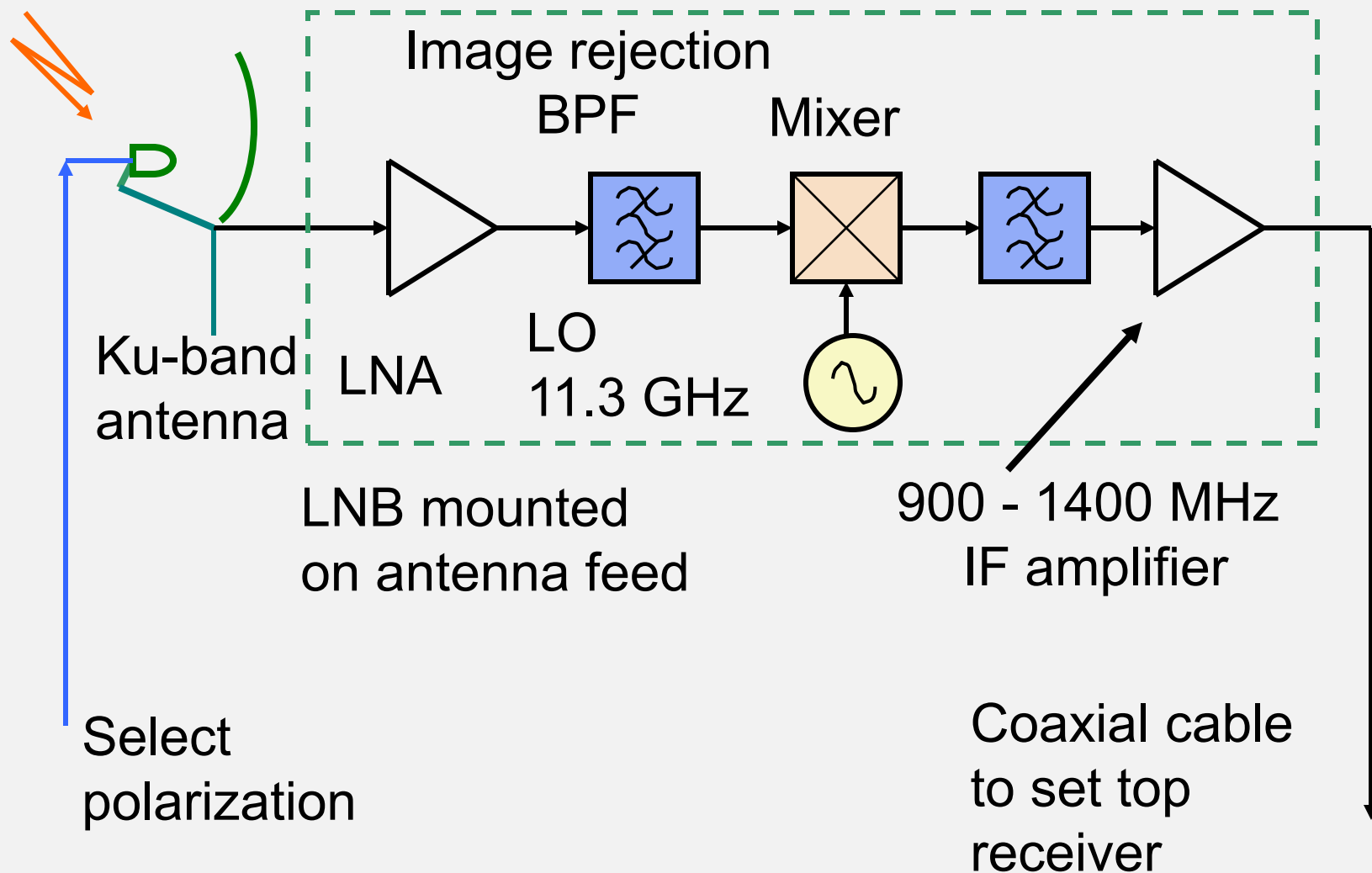
DBS-TV Receiving Stations

- 18 inch to 24 inch Ku-band dish antennas –
Additional features:
- Dual circular polarization - electronic switching
- 100 K LNB (Low Noise Block converter) –
usually mounted behind the antenna feed



Polarization Selection

- The signal comes in two flavors, Left Handed Circular Polarization (LHCP) and Right Handed Circular Polarization (RHCP)
- Depends on the sense of rotation of the electric field vector of the wave (polarization)
- Received simultaneously, a *polarizer* separates the two orthogonal polarizations for processing by separate LNBS
- Doubles the capacity of a system

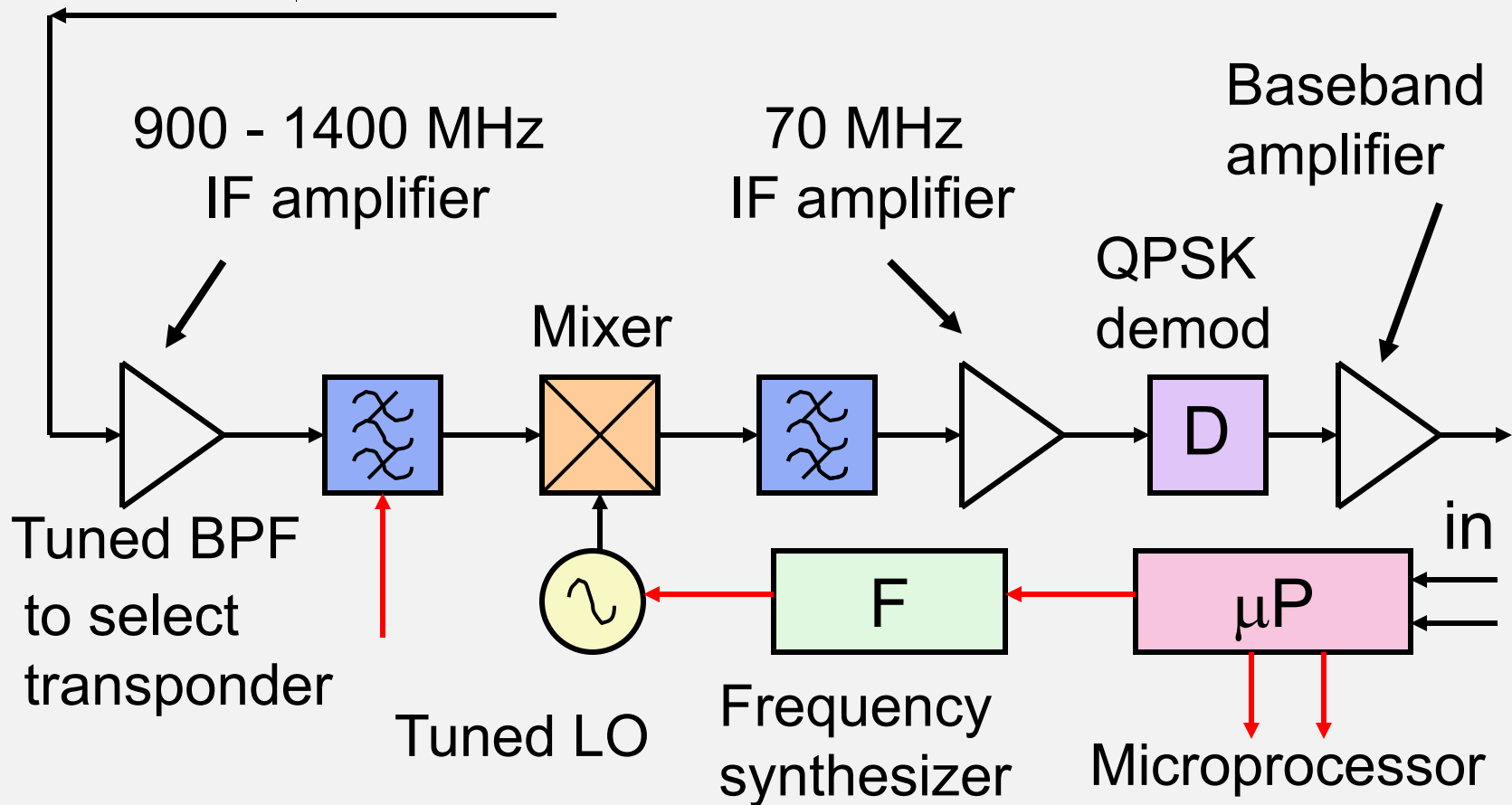


DBS-TV Receiver front end

Low Noise Block Converter (LNB)

- The low noise block converter (LNB) combines the LNA and downconverter (mixer + local oscillator)
- Note that the LNB is mounted directly to the output of the antenna
- Effectively suppresses the contribution to the system noise temperature of the (potentially) long run of cable that follows
- Allows for use of inexpensive cable

DBS-TV Receiver IF section



Summary

- Fixed Satellite Service (FSS) such as Intelsat uses large antennas, high power transmitters
- C/N on uplink (in transponder) is high
- Broadcast systems (such as DBS-TV) use small receiving antennas
- Satellite has high power HPA, large antennas
- Direct broadcast satellite TV has been highly successful but is ceding ground to cable and streaming services (internet)

Course Project: Preliminary (1/3)

- An important component of this course is the Course Project (CP) – 25% grade value
- Project teams will submit two reports and will be required to make a presentation
- A general introduction is provided in a document now posted to the Canvas website
- Adjustments may be made to lectures and HW dates to accommodate CP items

Course Project: Preliminary (2/3)

- First item: form a 3-person team and select a satellite or satellite system to analyze
- This information is due to the instructor by 11:59 PM on Wednesday, April 1
- A list of possible systems is included in the 'CP – Introduction' document
- Or you can propose your own

Course Project: Preliminary (3/3)

- The instructor will review the requests and may make changes
- E.g., add members or change the satellite system
- Team membership and topic should be settled by noon on Thursday, April 2
- The first written report will be due by the start of class on Wednesday, April 8 (maximum length: 3 pages of text, details to follow)

Modulation

- Radio frequency transmission requires a *carrier wave*
- The carrier frequency defines the frequency of the radio transmission
- E.g., 89.1 MHz is the carrier frequency of WVTF in Roanoke, VA
- Question: does a pure sine wave carry any information?

Modulation

- Answer: A pure sine wave only carries information about its existence and its defining parameters (such as frequency)
- We must *modulate* the carrier wave to send useful information

Modulation

- The carrier is described by the equation

$$V(t) = A_c \cos (\omega t + \phi)$$

where:

A_c is the carrier amplitude

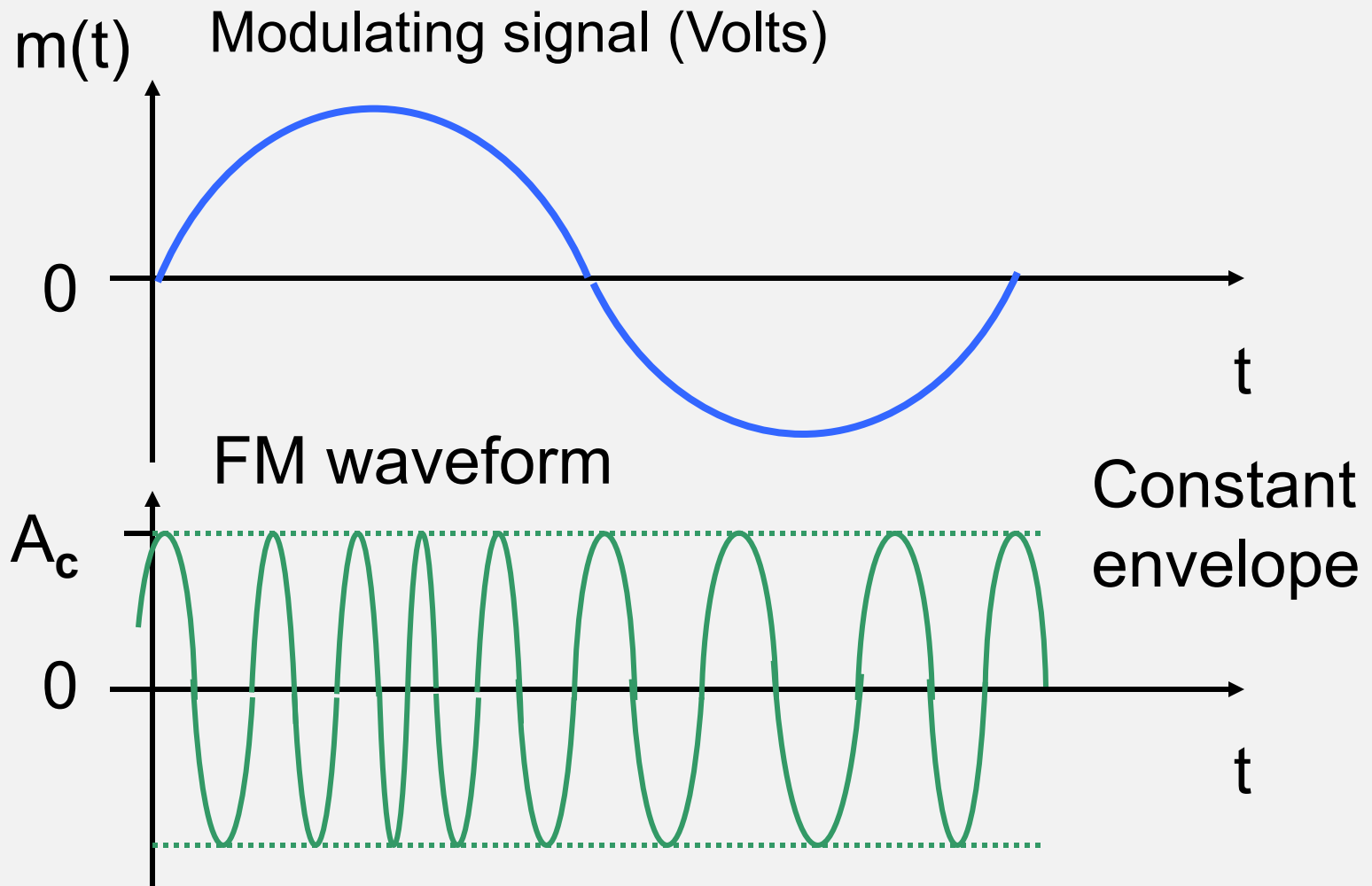
ω is the angular frequency = $2\pi f$ [rad/s]

ϕ is the phase angle of the carrier [rad]

- We can modulate any of the three parameters of the carrier to carry information (*baseband*)

Modulation

- Modulate A_c = Amplitude Modulation (AM)
- Modulate ω = Frequency Modulation (FM)
- Modulate ϕ = Phase Modulation (Ph M)
- If the modulating parameter varies in a continuous manner then the signal is *analog*
- For the first 20 years of satellite communications analog signals were widely used with frequency modulation (FM)



FM waveform generated with sine wave modulation

Analog / FM modulation

- Satellite communications began with analog signals and FM modulation
- AM modulation was always considered unsuitable because of varying signal power
- *Wideband* FM was able to operate at low C/N ratios (5 – 10 dB)
- The penalty was the need for large bandwidths
- Analog methods are now obsolete for commercial satellite communications

Digital Transmission of Analog Signals

- With a digital signal the modulating parameter takes on discrete values
- Most analog signals are now converted to digital form for sending
 - Telephones
 - Satellite Television, HDTV
 - Digital Cable TV
 - Streaming services
- Analog signals are often stored in digital form

Digital TV transition

- TV began as an over-the-air analog service (1930s - 1940s)
- In the early 2000s the U.S. government mandated a switchover to digital broadcasting
- Full-power stations transitioned on June 12, 2009
- Converter boxes were offered at a discount to ease the pain to consumers
- The last low-power stations (in Alaska) were required to convert by January 10, 2022

Analog Telephone retirement

- Similar to TV, telephone service was analog when introduced in the late 1800s
- Based on twisted pairs of copper wire
- At its peak, the penetration of landline analog telephones into U.S. homes was nearly 100%..
- ..now down to a small percentage, and costly to maintain

Analog Telephone retirement

- The FCC is directing the phase-out of *plain old telephone service* (POTS)
- Running up against resistance from older folks
- It seems the FCC has given up on mandating the end of POTS lines
- Instead, it deregulated pricing for POTS effective August 2, 2022
- The price for POTS might skyrocket as a result

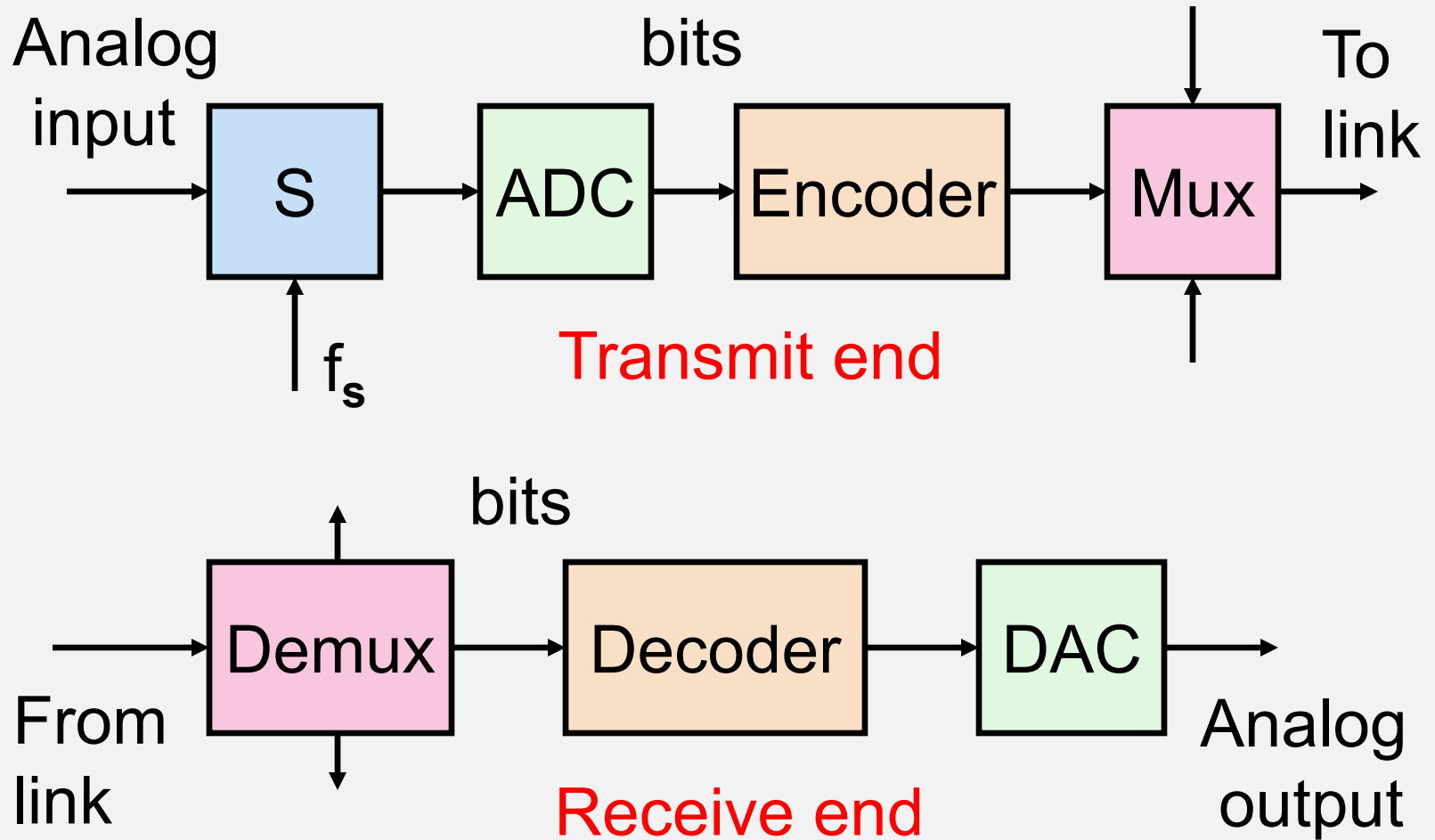
Digital Transmission of Analog Signals

- Advantages of digital transmission:
 - Digital circuits are more reliable than analog circuits
 - Errors can be detected and corrected
 - Digital switching can be used
 - Different signals can be *multiplexed* easily with many other types of signal
 - Quality of output baseband signal can be better

Digital Transmission of Analog Signals

- Several steps are needed to set up on the TX side:
 - Sample analog signal (*sampling*)
 - Convert samples to digital words (*quantizing*)
 - Multiplex signal with other bit streams (*encoding*)
- Then transmit bits over link

Digital transmission of analog signals



Digital Reception of Analog Signals

- On the RX side after receiving the signal:
 - *Demultiplex* to obtain wanted signal
 - Detect and correct bit errors at receiver
 - Convert digital words to analog signal

Sampling Theorem

- Can we recover an analog signal from samples without introducing noise or distortion?
- The answer is yes!
- Harry Nyquist published a paper in the AIEE Proc. in 1928
- Now known as *Nyquist's Sampling Theorem*
- See Appendix D for a wider discussion



Harry Nyquist (1889 - 1976)

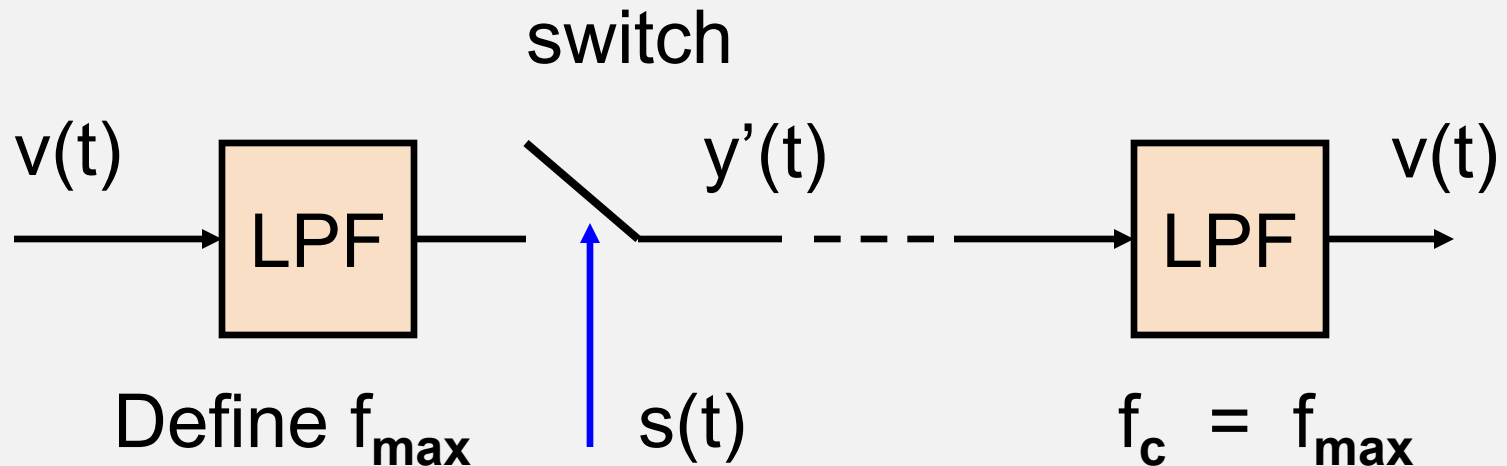
Sampling Theorem

- Nyquist's Sampling Theorem in words:

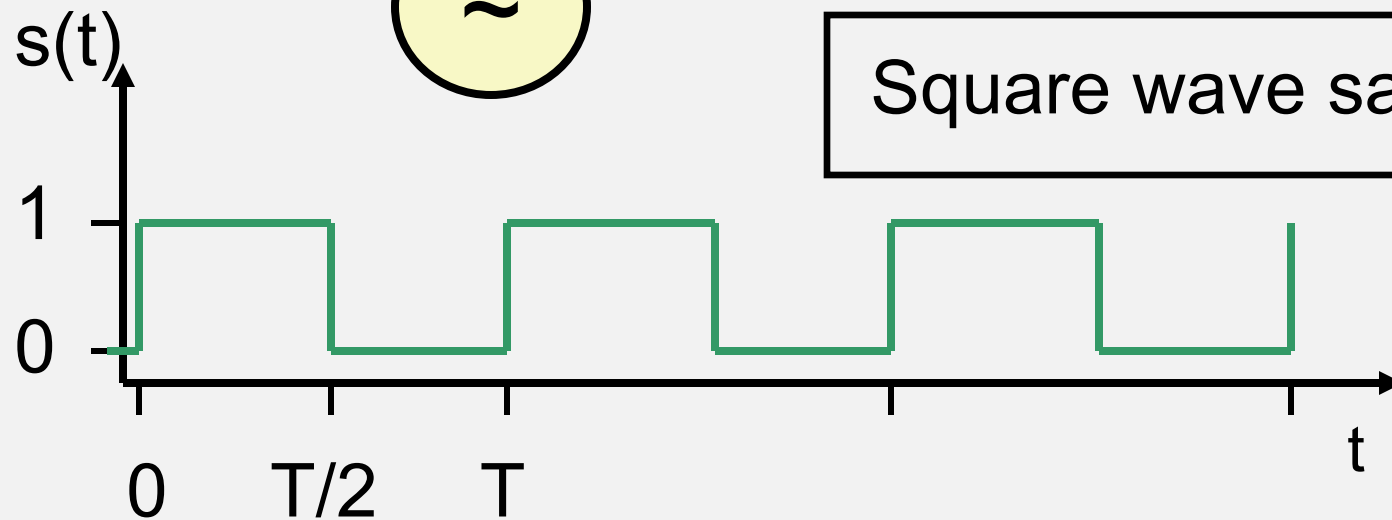
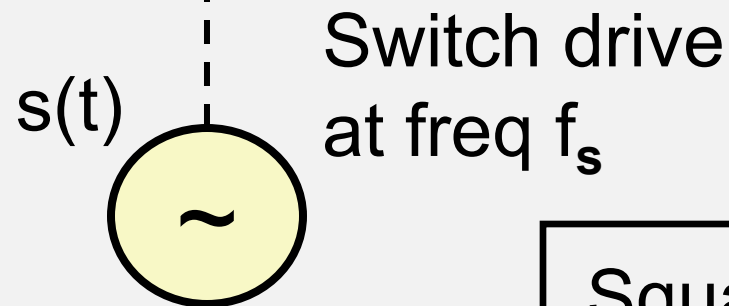
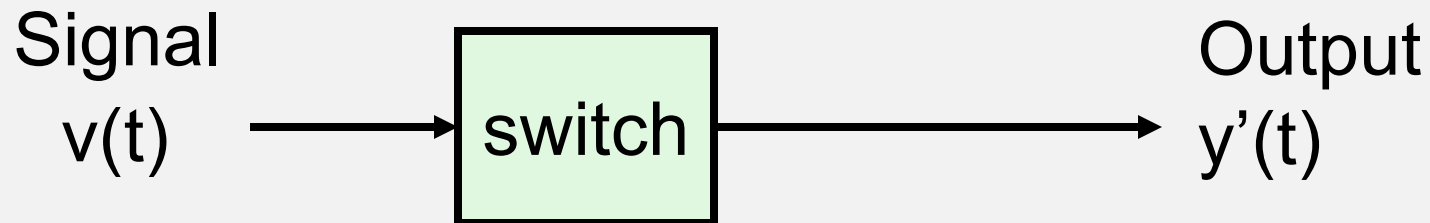
Any analog signal can be completely recovered without distortion from a series of uniformly spaced samples of the signal, provided that the samples are made at a rate exceeding twice the highest frequency present in the signal

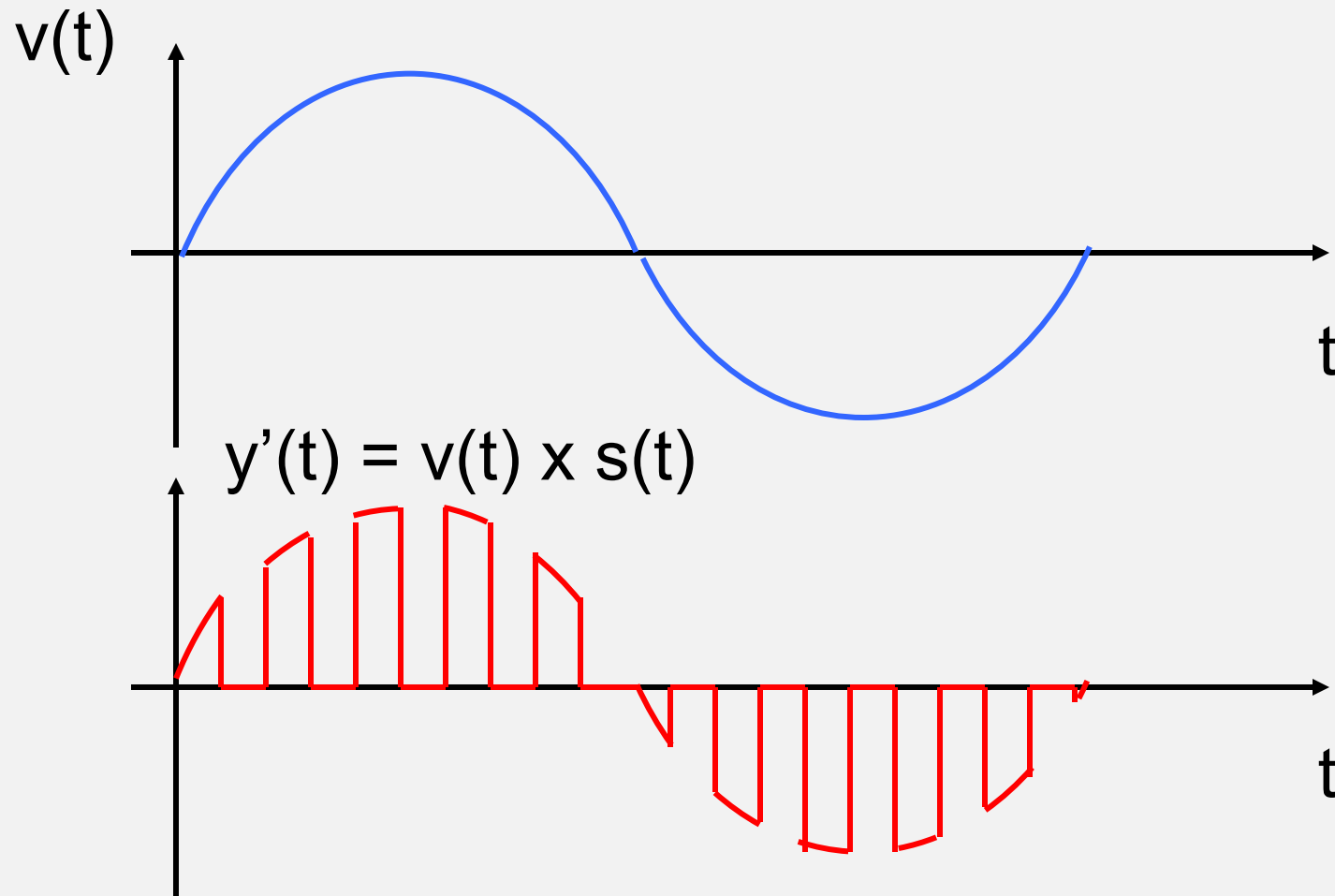
- * The frequency range of the signal is always controlled by filtering

Sampling Theorem



Sampling and recovery of an analog signal.
 $v(t)$ is multiplied by 1 when switch is closed,
0 when switch is open





Waveform generated by square wave sampling switch

Time waveforms and frequency spectra

- Communications engineers work with frequency spectra
- Systems generate time waveforms
- We use the *Fourier Transform* (FT) or *Fourier Series* to move between the time and frequency domains
- We rarely calculate FTs - we use known FT forms to show where energy lies in the frequency spectrum

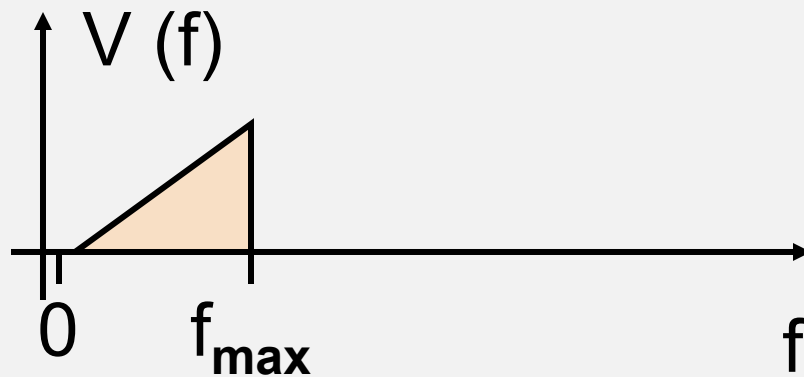
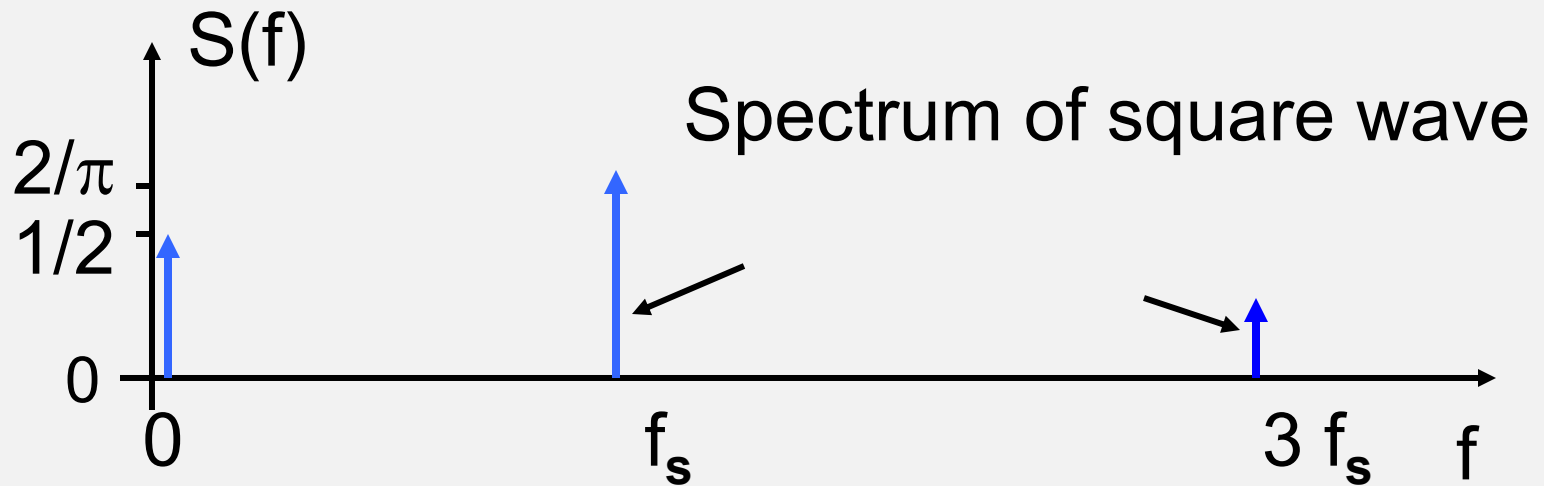
Time waveforms and frequency spectra

- Fourier Series for a square wave with period T seconds and voltage levels 0 and 1 V

$$s(t) = \frac{1}{2} + \frac{2}{\pi} [\cos \omega_o t - \frac{1}{3} \cos 3\omega_o t + \dots]$$

where $\omega_o = 1 / 2\pi T$ or $f_o = 1 / T$

- Square wave has DC component = $\frac{1}{2} V$
- Fundamental frequency f_o with magnitude $2/\pi$
- All odd harmonics with amplitudes $2 / n\pi$
where $n = 1, 3, 5 \dots$



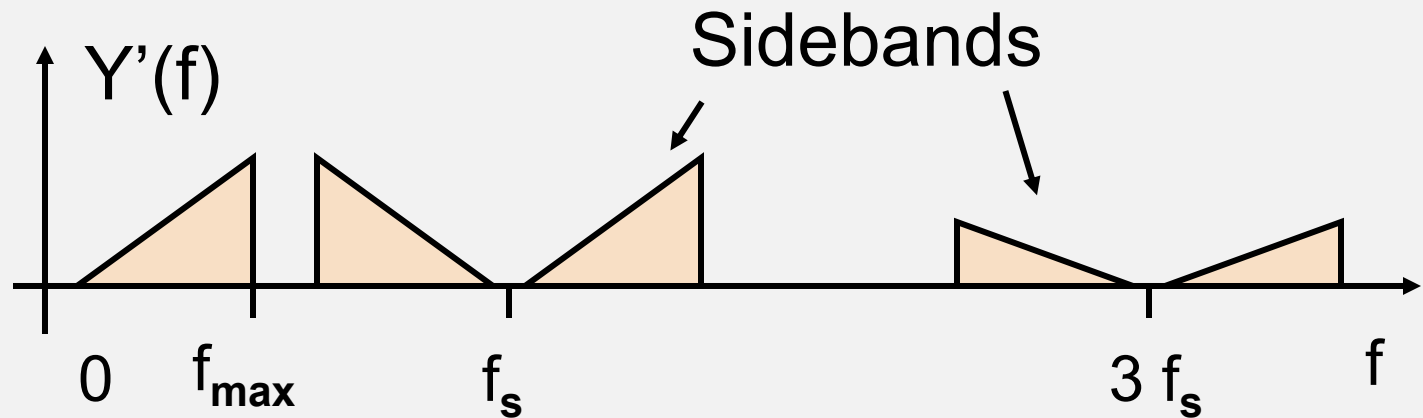
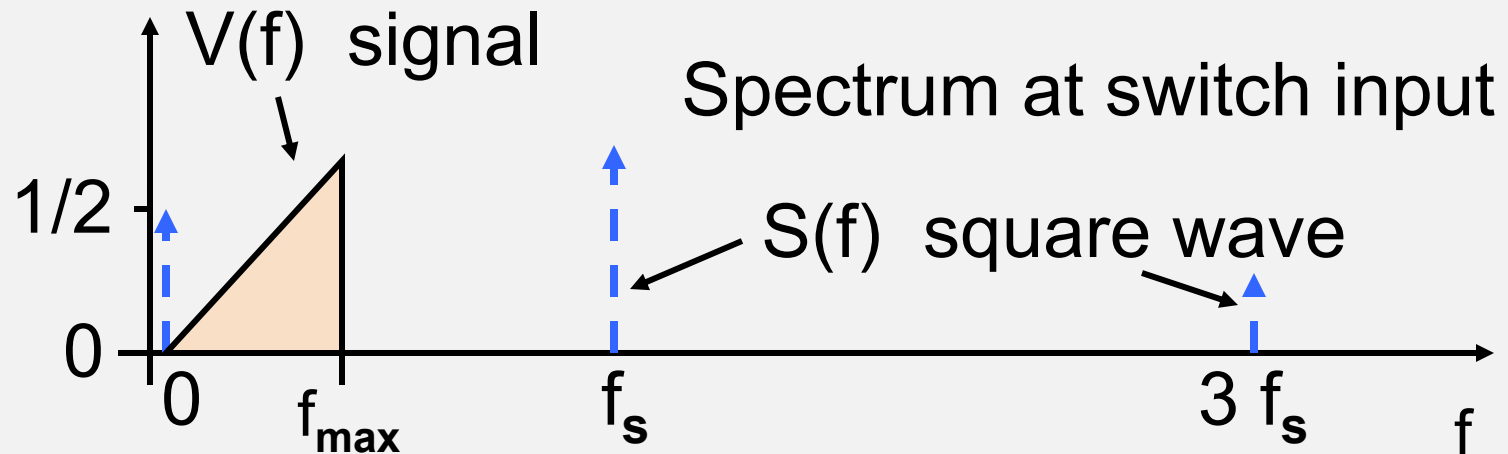
Spectrum of arbitrary baseband signal $v(t)$

Sampling

- We assume an arbitrary input $v(t)$ with a spectrum that extends to $f_{\max}$ (probably limited by a filter)
- Sampling process multiplies signal $v(t)$ by square wave $s(t)$

$$y(t) = v(t) \times s(t)$$

- In frequency space this is equivalent to a *convolution* of the spectra of $v(t)$ and $s(t)$



Spectrum at switch output

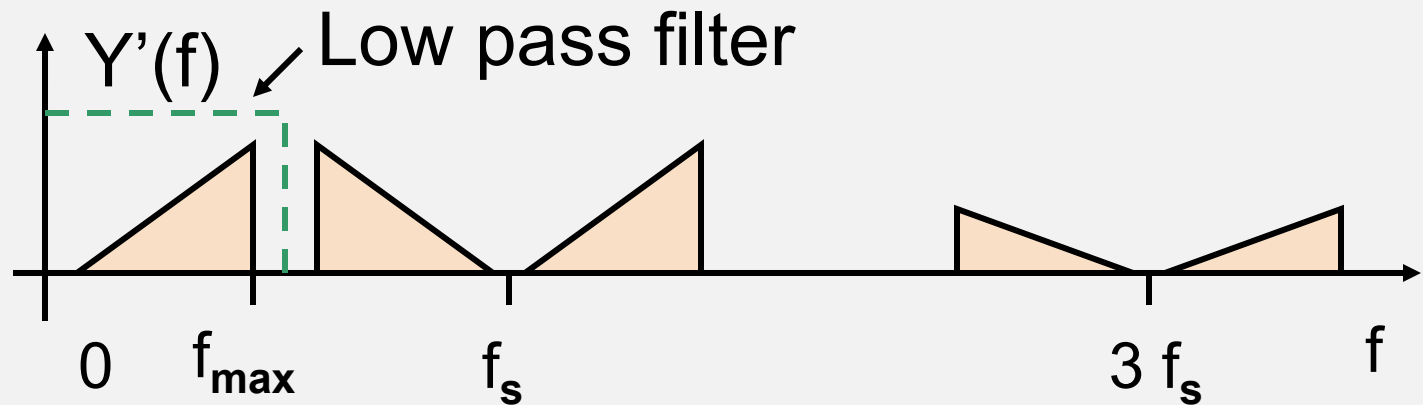
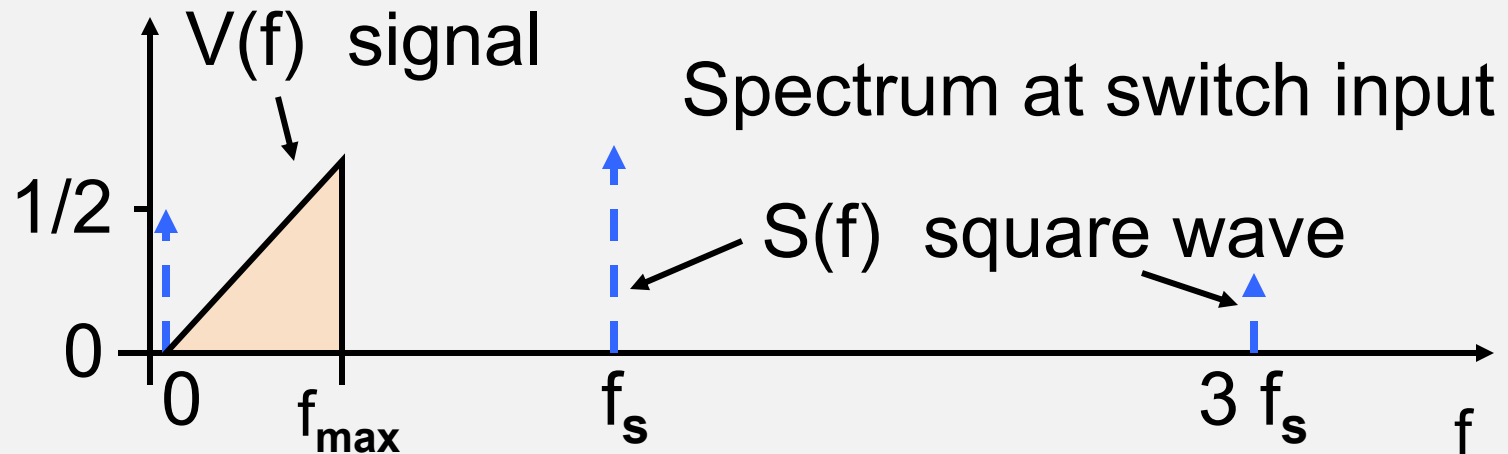
Spectrum at output is convolution of input spectra

Sampling

- In the time domain we have $y(t) = v(t) \times s(t)$
- Hence

$$y(t) = \frac{1}{2} v(t) + v(t) \frac{2}{\pi} [\cos \omega_o t - \frac{1}{3} \cos 3\omega_o t + \dots]$$

- The original signal is present in the first term $\frac{1}{2} v(t)$
- We can recover the $v(t)$ by removing all the other terms with a low pass filter

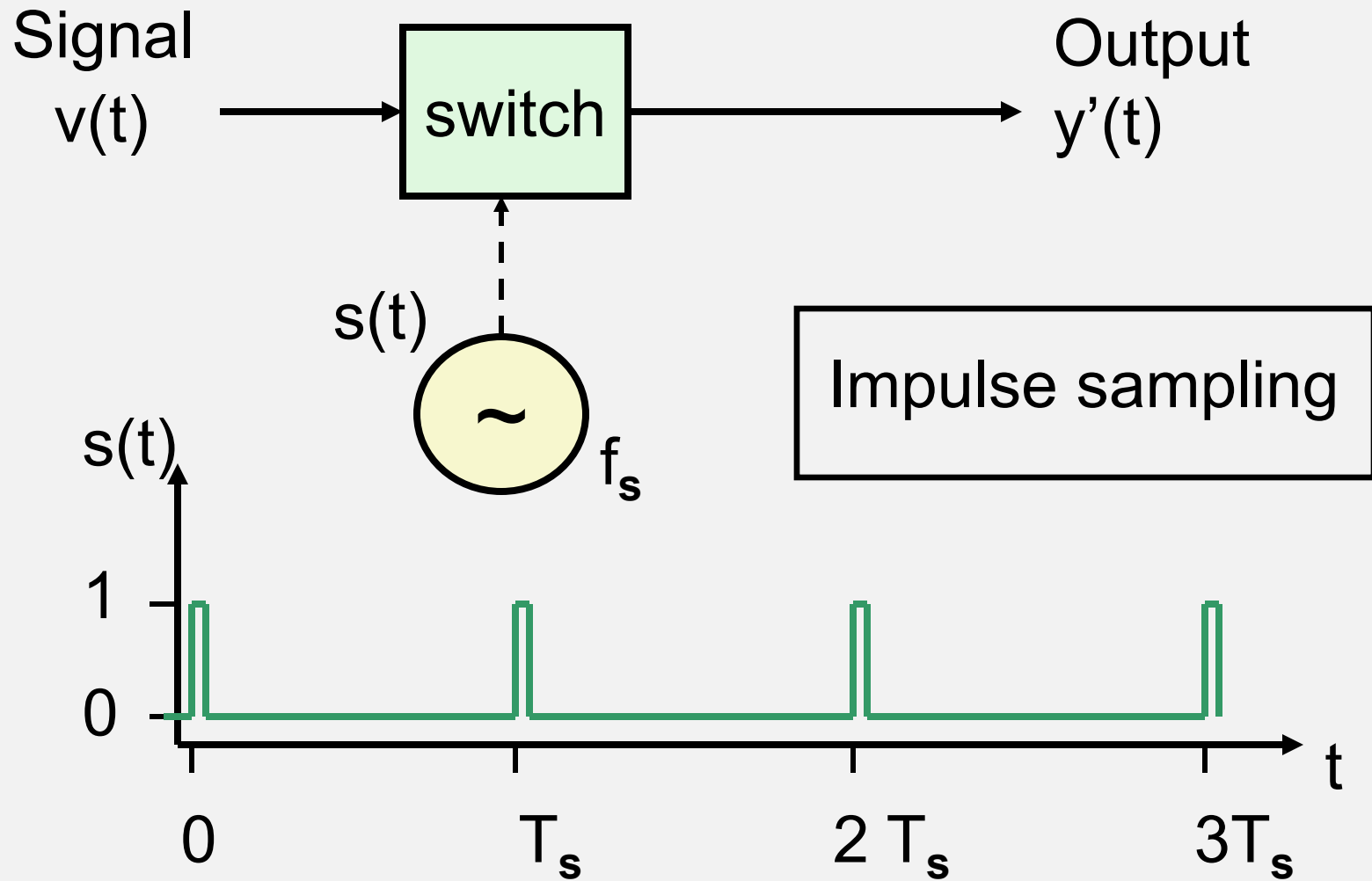


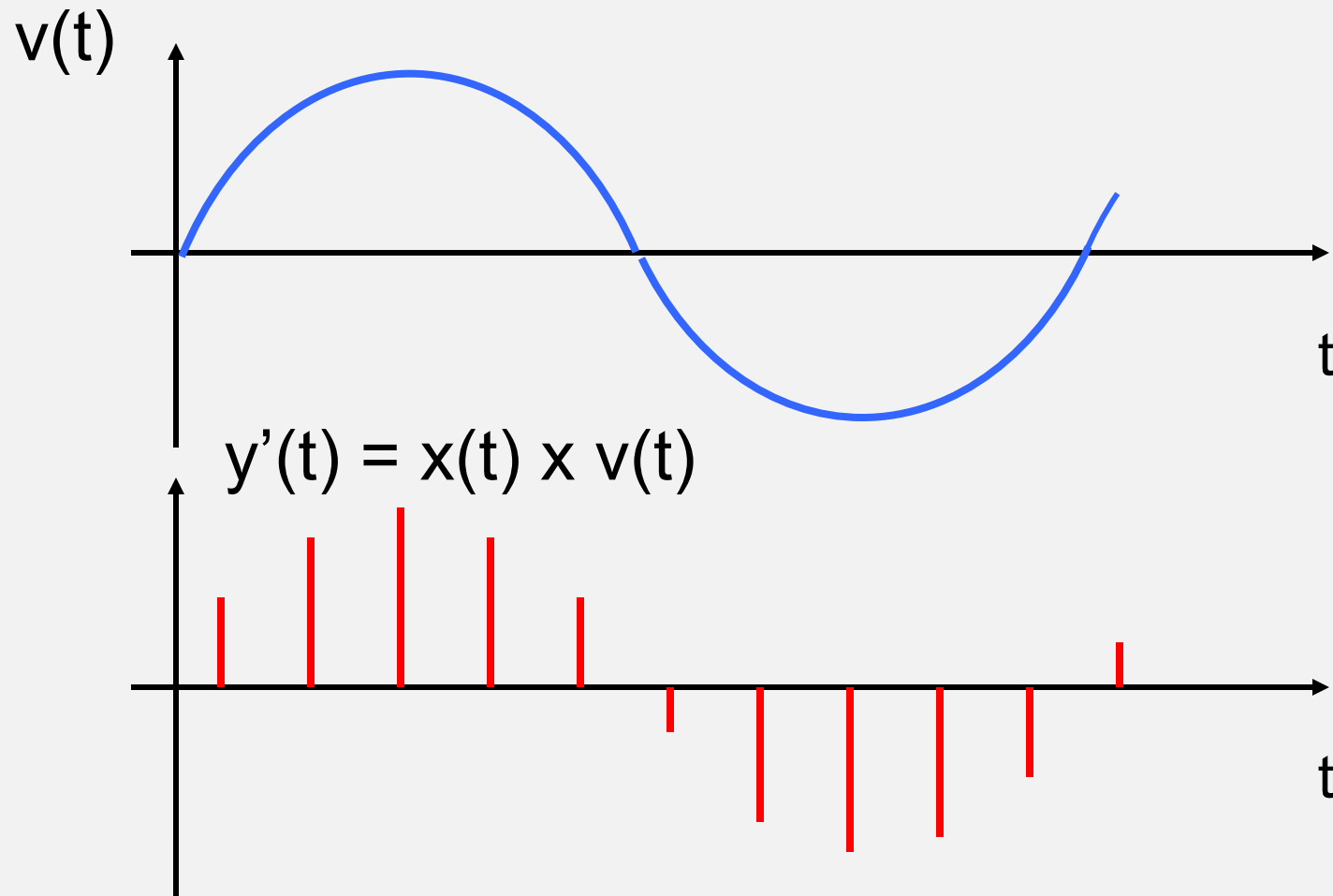
Spectrum at switch output

Spectra in sampling and recovery processes

Sampling with Impulses

- The sampling theorem does not require the switch to be driven by a square wave
- We can use short sampling pulses
- Introduces additional spectral components
- In the limit, short pulses $\Rightarrow$ impulses
- Baseband signal component is still present
- Short pulses make multiplexing easier

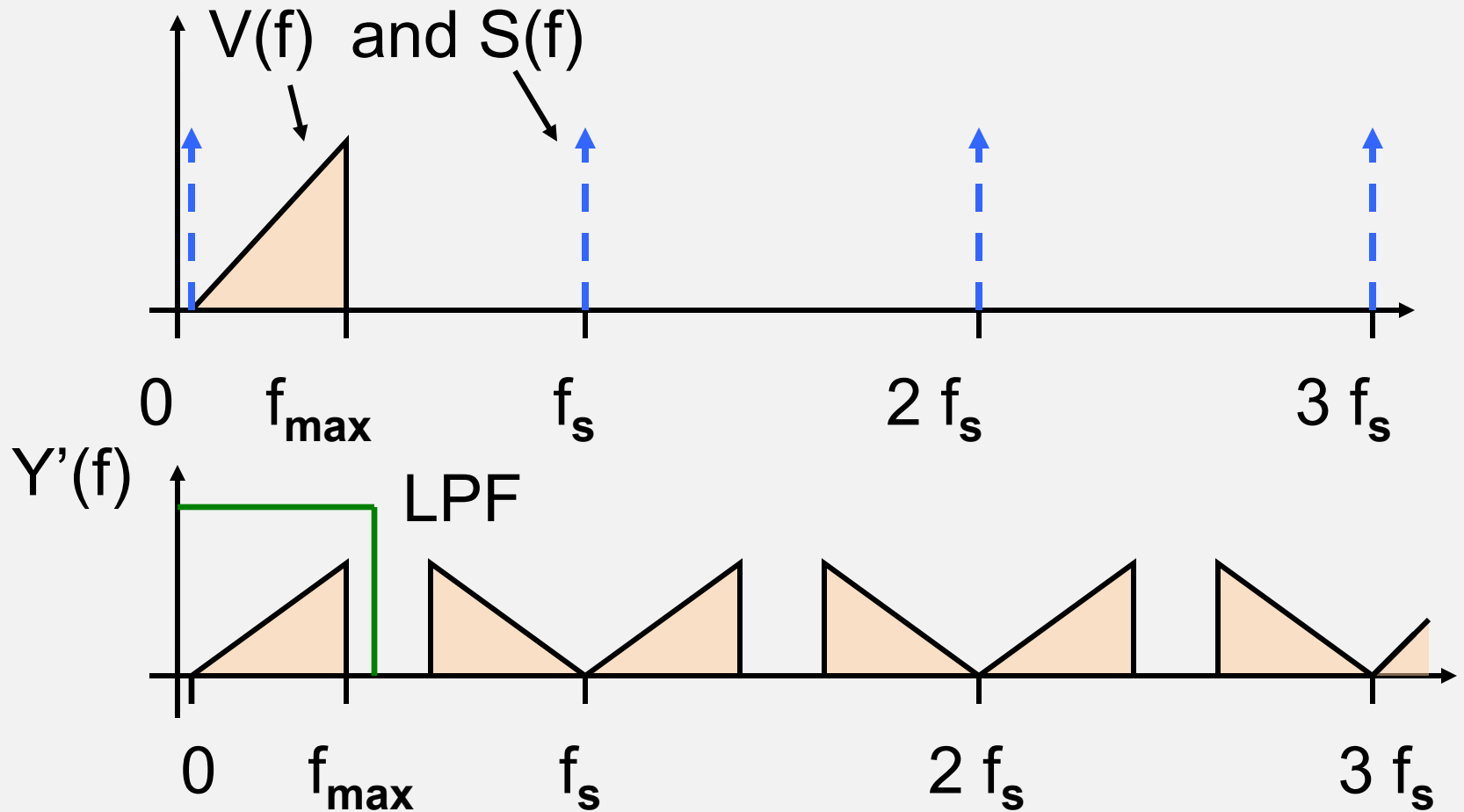




Waveform generated by impulse sampling

Sampling with Impulses

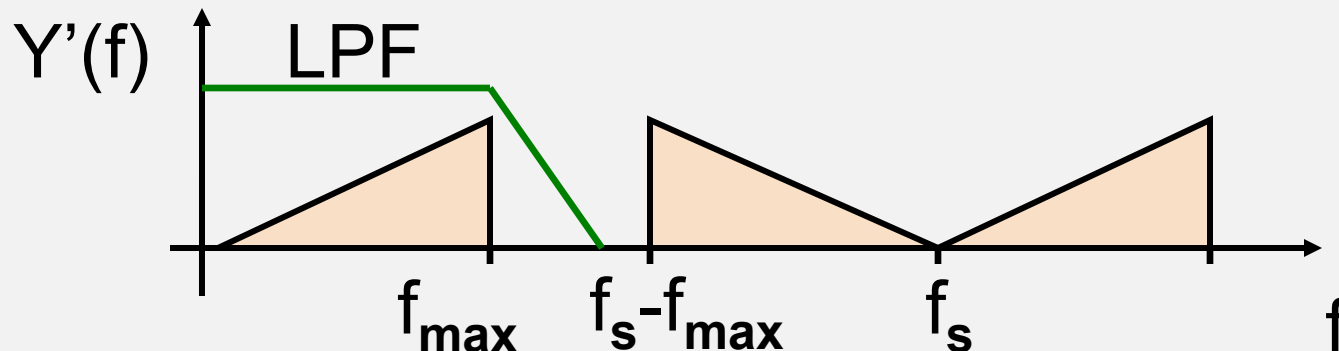
- The spectrum of an infinite series of pulses in the time domain is an infinite series of pulses in the frequency domain with separation $f_s = 1 / T_s$
- Applying $y(t) = v(t) \times s(t)$ is again equivalent to convolving the spectra of $v(t)$ and $s(t)$
- The spectrum of baseband is reproduced at every pulse in the frequency domain



Spectra for impulse sampling

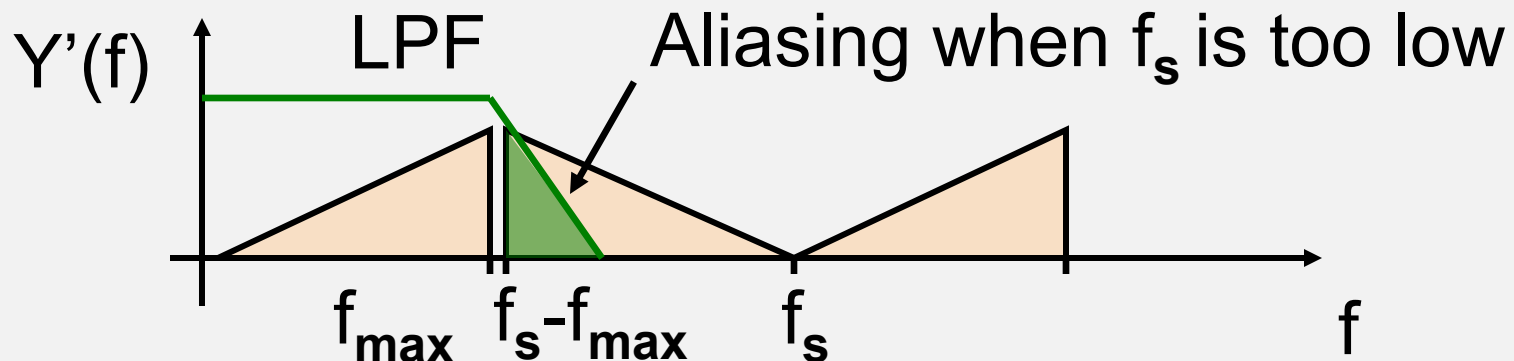
Recovering Analog Signals

- We can recover the original signal from the sampled signal with a low pass filter
- Require that $f_s - f_{\max} \geq f_{\max}$ (need separation)
- Hence $f_s \geq 2 f_{\max}$
- If LPF is not ideal (always true) $f_s > 2 f_{\max}$



Aliasing

- If we fail to meet the requirement that $f_s - f_{\max} \geq f_{\max}$, i.e., the sampling frequency is too low, we get *aliasing*
- Part of lower sideband at f_s gets into LPF
- Aliasing introduces noise into recovered signal



Summary

- Sampling Theorem allows accurate recovery of analog signals sent as samples
- Sampling rate must meet Nyquist's criterion
- For real filters, make $f_s > 2 f_{\max}$
- Aliasing occurs if f_s is too low
- In telephony $f_s = 8 \text{ kHz} = 2.35 \times 2 f_{\max}$
- $f_s = 2 f_{\max}$ is called *Nyquist rate* sampling

Analog to Digital Conversion

- First step in digital transmission of analog signal is sampling at a high enough sampling rate
- Next step in is to create a digital word from the samples
- Number of bits in the word (N) is important
- Digital words have discrete values
- The word is a *quantized* version of sample

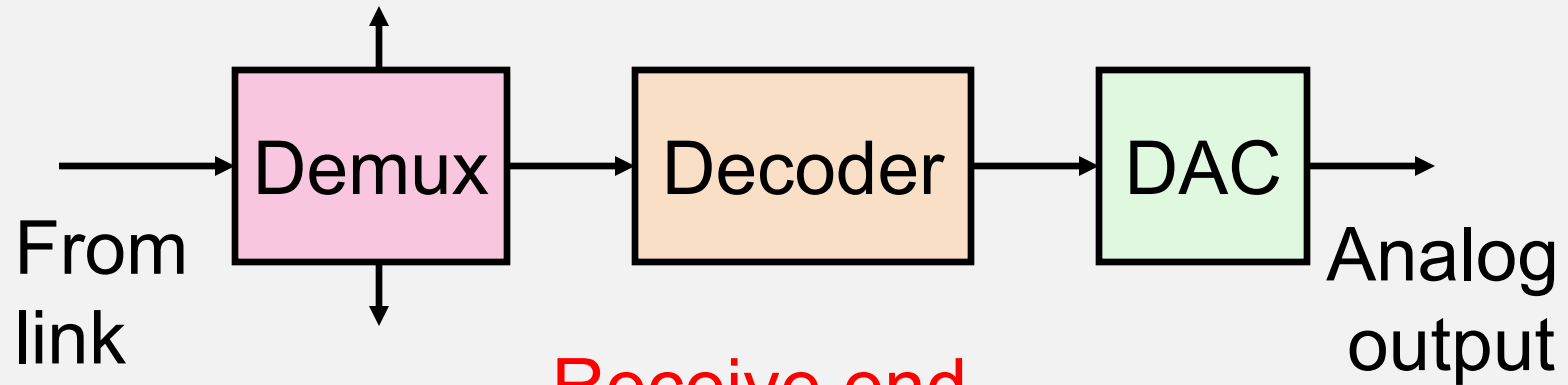
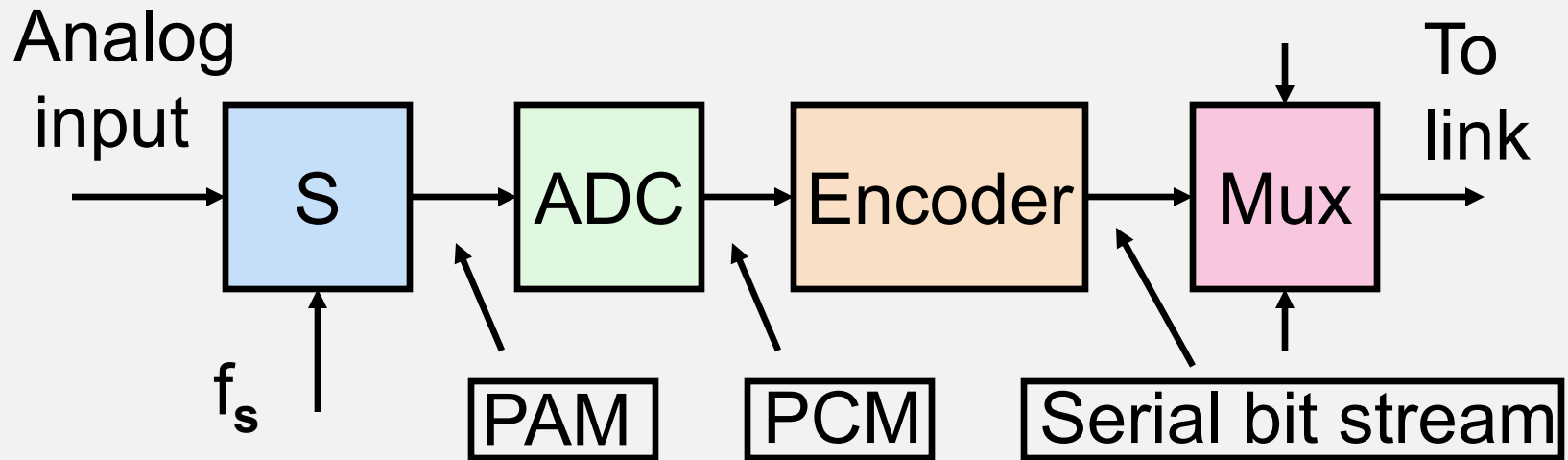
Analog to Digital Conversion

- Analog output of sampling process is called *Pulse Amplitude Modulation (PAM)*
- With PAM, the samples can have any value
- Digital encoding of samples with quantization is called *Pulse Code Modulation (PCM)*
- With PCM, the encoded samples can have only discrete values as determined by the quantization scheme

Pulse Code Modulation: PCM

- PCM has pulses
- It is not really a code
- It is not really a modulation
- PCM is an obsolete term in widespread use
- Usually refers to voice signals converted to bits
- Alternatives are *digital voice*, *digital video*

Transmit end

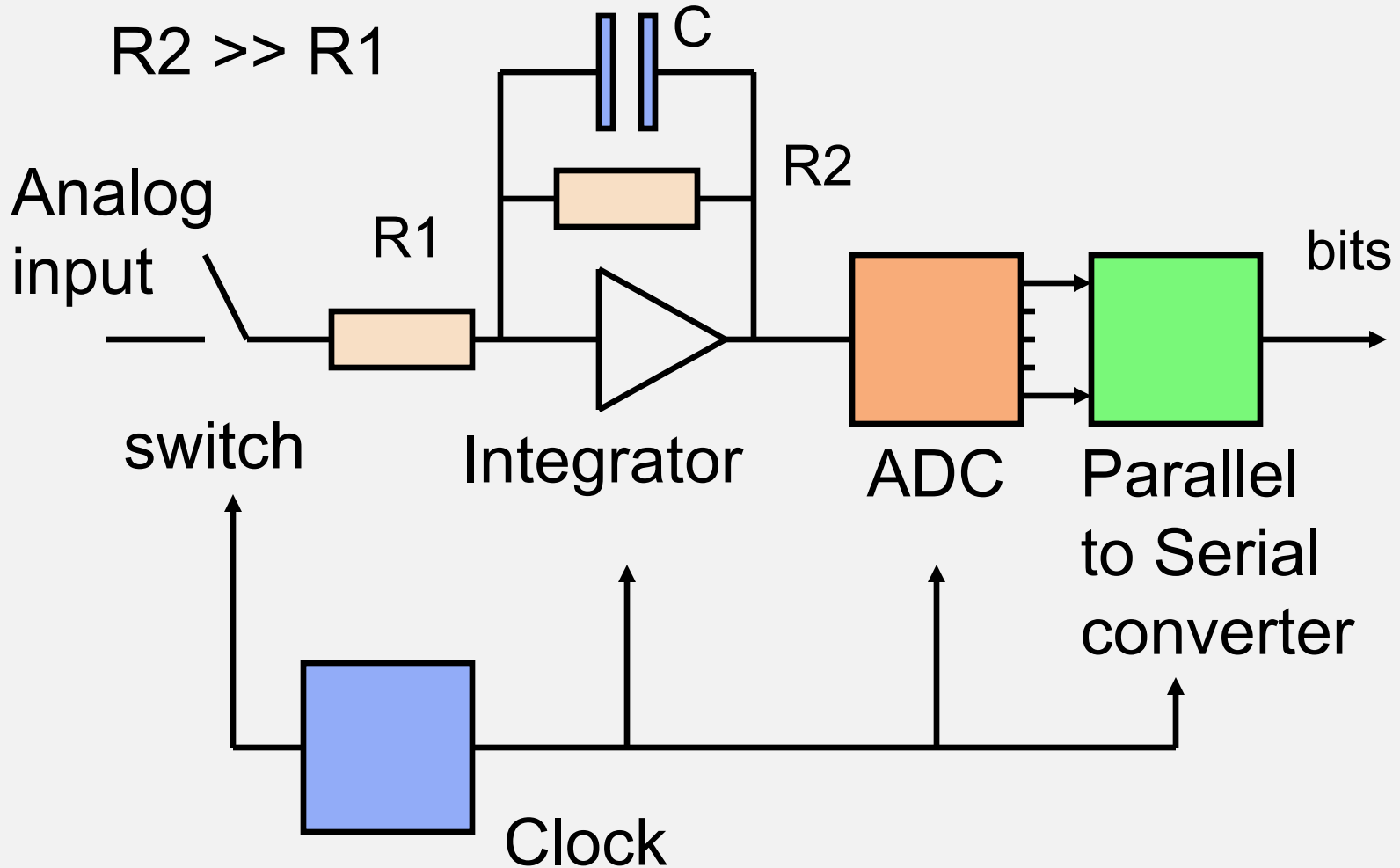


Receive end

Analog to Digital Conversion

- Samples are made with *Sample and Hold* circuit
- Fast charge / slow discharge integrator
- ADC is a specialized circuit that outputs a digital word corresponding to sample value
- ADCs are characterized by speed and number of bits

Sample and Hold and ADC



Bit Rates

- Suppose we digitize to N bits per sample
- The bit rate at the output of the parallel to serial converter is r_b where

$$r_b = f_s \times N$$

- Bit rate sets the bandwidth required of the communication channel
- More bits gives better S/N ratio but requires more bandwidth

Bit Rates

- Example #1:
- Sample rate for telephony is 8 kHz

$$\text{Bits/sample} = N = 8$$

$$\text{Transmission rate} = f_s \times N = 64 \text{ kbps}$$

- International standard of all telephone traffic

Bit Rates

- Example #2

Sample rate for CD recording = 44 kHz

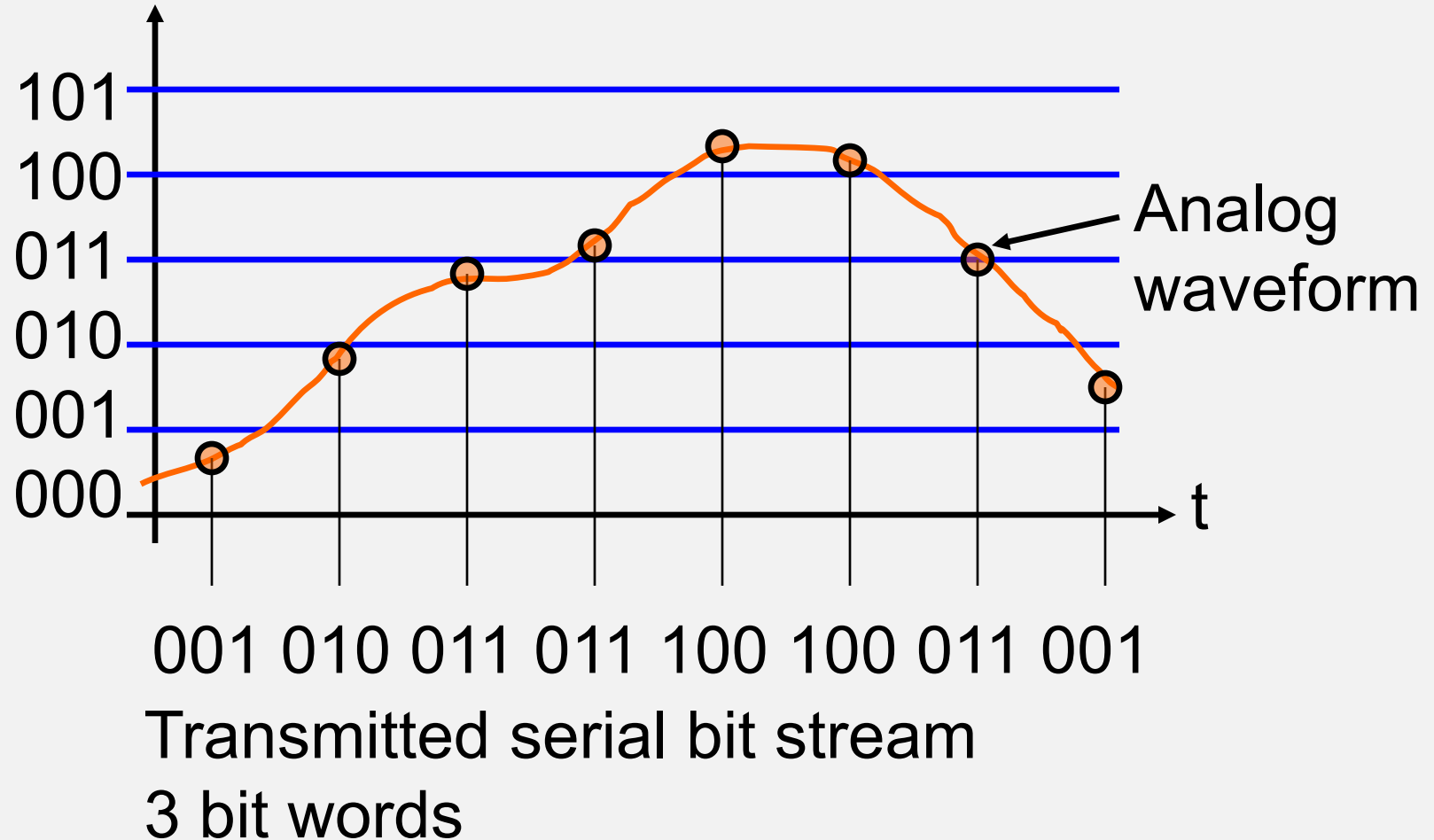
$N = 16$ bits, $r_b = 704$ kbps / channel

- CD records two channels (stereo) in encoded form
- *Message data rate* is 1408 kbps
- Actual bit rate of recording is ~ 3.6 Mbps
- Difference is in coding, packetizing, etc

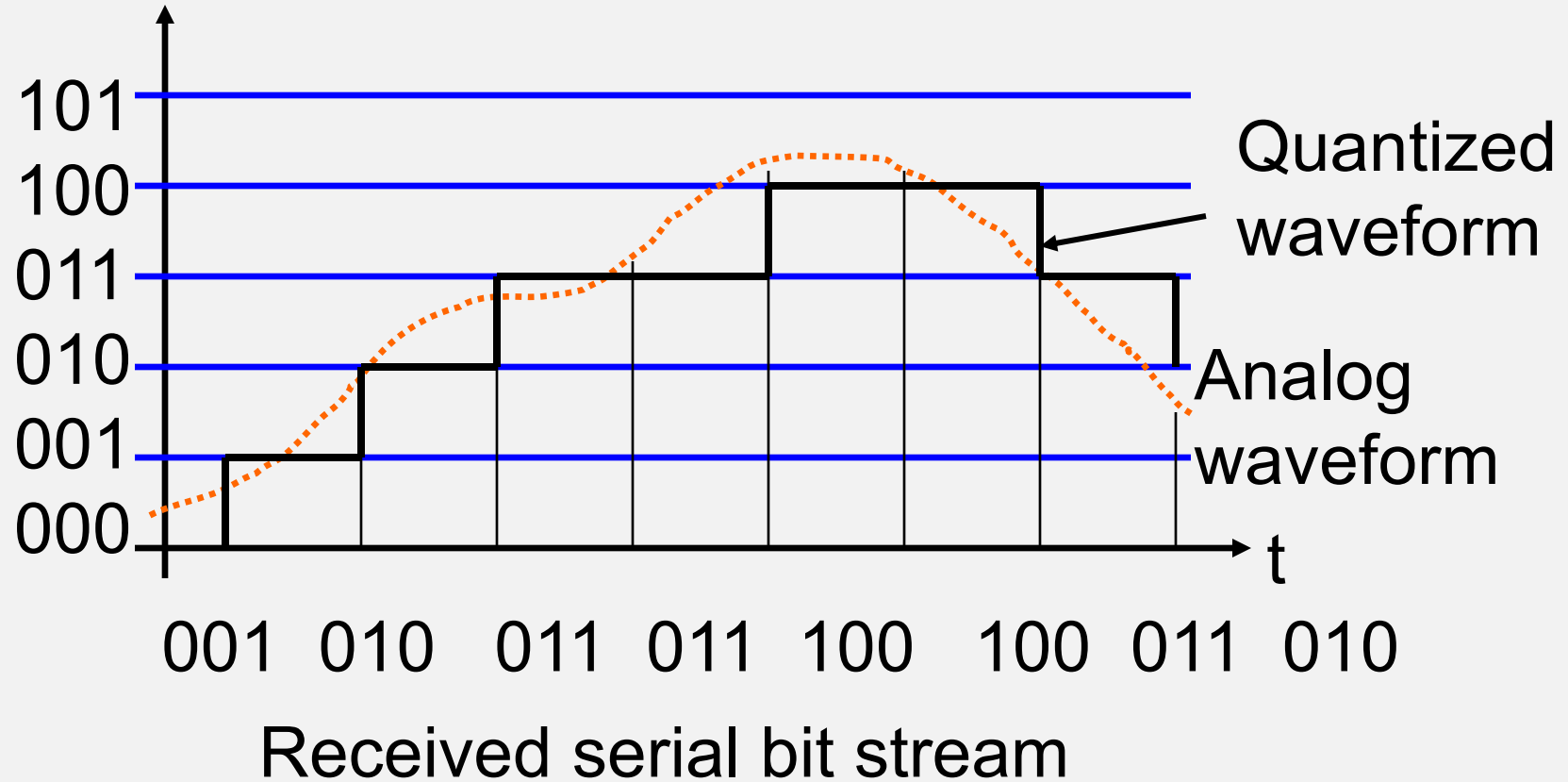
Quantization

- The ADC converts a voltage sample to a digital word
- Each digital word corresponds to a specific voltage level at the analog input to the ADC
- There are $2^N - 1$ voltage levels
- The ADC outputs the word that has a voltage level closest to the input signal
- This process is called quantization

Six level quantization



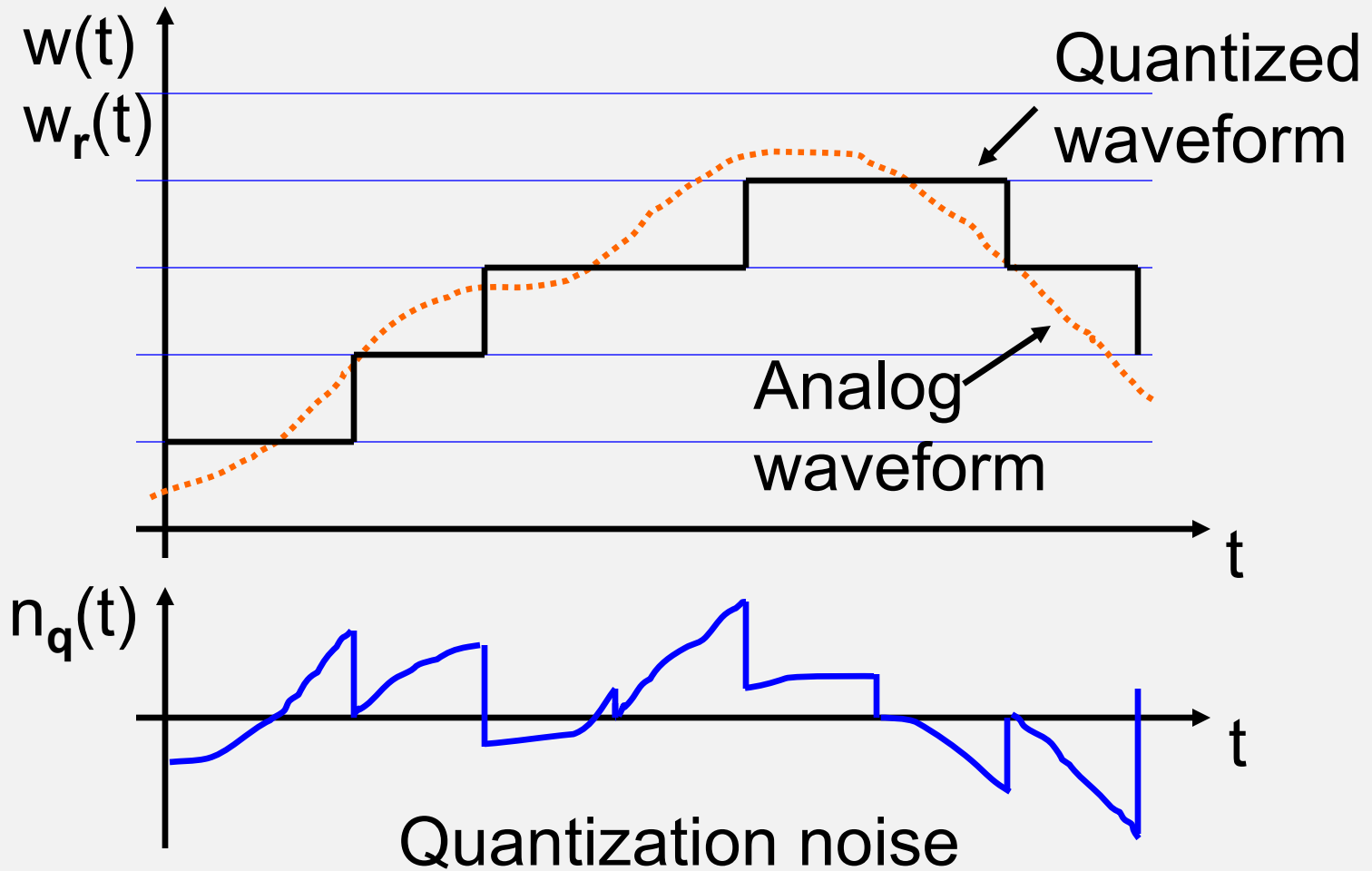
Receiver waveform with 3 bit words



Quantization Noise

- The quantized waveform is different from the analog waveform
- There is an error because the quantized waveform, $w_r(t)$, is not the same as the original signal $w(t)$
- We describe the error as $n_q(t)$ where
$$w_r(t) = w(t) + n_q(t)$$
- $n_q(t)$ is called *quantization noise*

Quantization noise with 3 bit words



Quantization Noise

- Quantization noise adds to the recovered analog signal and generates noise in the baseband signal (what is heard, seen)
- For uniform quantization and no bias in voltage level, baseband signal to (quantization) noise ratio is

$$(S/N)_Q \approx 6 N \text{ dB}$$

- E.g., for linear quantization with 8 bits

$$(S/N)_Q \approx 6 \times 8 = 48 \text{ dB}$$

Summary: Analog to Digital Conversion

- Many sources of signal are inherently analog - digital communication techniques require an ADC followed by a DAC
- PCM entails digital encoding of signals with quantization
- High bits rates from digitization require large bandwidths
- Quantization noise generates noise in the baseband signal

